



Information Retrieval:

Searching with Formulas and Text, and a Framework for Characterizing Search by Humans & Systems

Richard Zanibbi

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Professor of Computer Science

Rochester Institute of Technology, USA

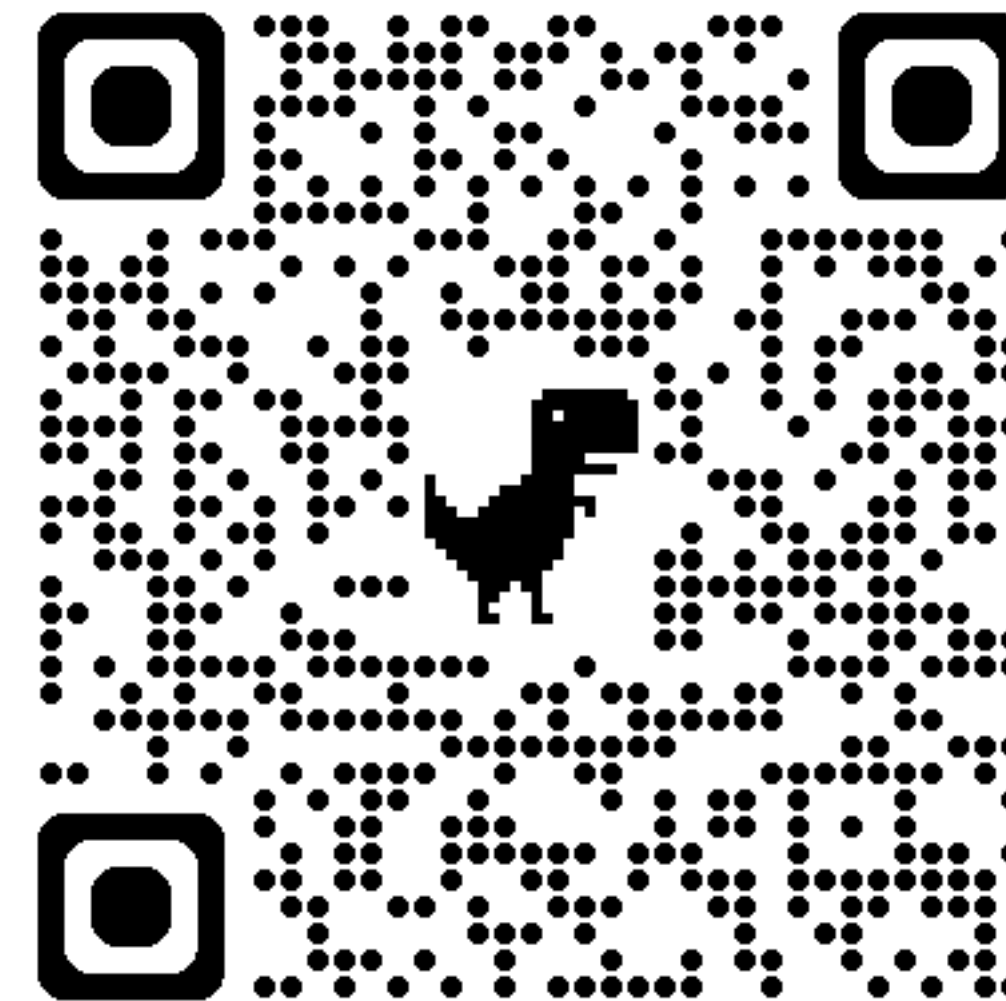
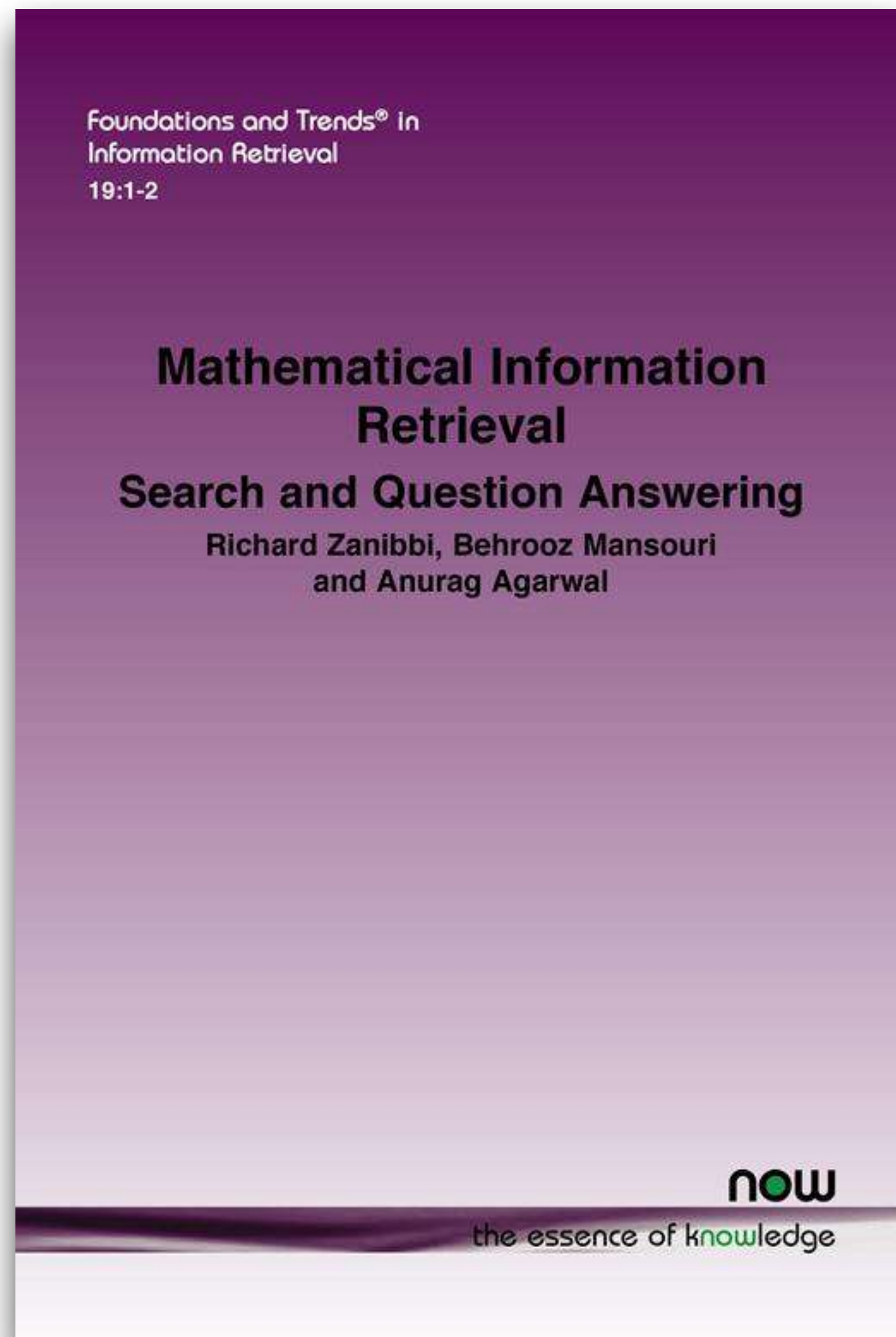
May 28, 2026



Mathematical Information Retrieval: Search and Question Answering

Richard Zanibbi, Behrooz Mansouri, and Anurag Agarwal

Foundations and Trends allowed the final PDF to be released for free! Link to PDF:



Interpreting & Representing Math Formulas

$$idf(t_i) = \log \frac{N}{n_i}$$

An Example Technical Document Excerpt

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Unless already known in the intended context, text and prior knowledge are *needed* to interpret formulas

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Taken from “Understanding Inverse Document Frequency” by Robertson (2004)

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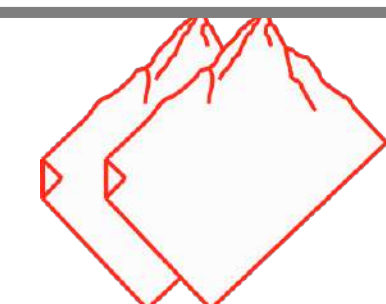
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Example 2.1: Inverse Document Frequency (*IDF*)

Excerpt from Robertson (2004)

...Assume there are N documents in the collection, and that term t_i occurs in n_i of them ... the measure proposed by Sparck Jones, as a weight to be applied to term t_i , is essentially

$$idf(t_i) = \log \frac{N}{n_i} \quad (1)$$



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Textual context / information

Variables

Functions and Operators

Additional Context



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Variables: placeholders for a set of values, similar to common nouns

- The text identifies N as the number of documents in a collection. N is like a common noun, because the collection is not specified.
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- $\log$: log function with unspecified base.
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- Division ($\frac{\quad}{\quad}$), application ($idf(\cdot)$, $\log \cdot$), and equivalence ($=$) appearing in Equation (1) that are not defined in the excerpt.

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Additional context:

- The text indicates Sparck Jones introduced the idf formula in a different, unspecified form (Jones, 1972).

Formulas in Text (Symbol Layout Trees)

Sequence of Tokens and Symbol Layout Trees

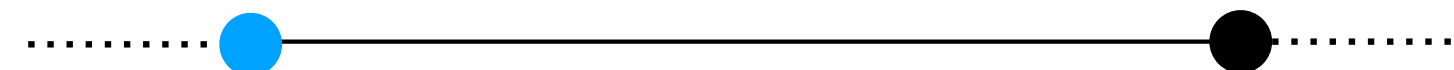
Example 2.5: Text tokens, formula tokens, and visual formula structure

Sequence of text (black) and formula (blue) tokens

Assume there are N documents in the collection, and that term t_i occurs in n_i



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Symbol Layout Tree (SLT)

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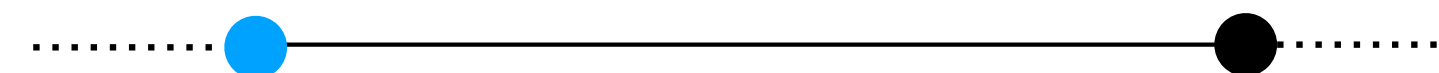
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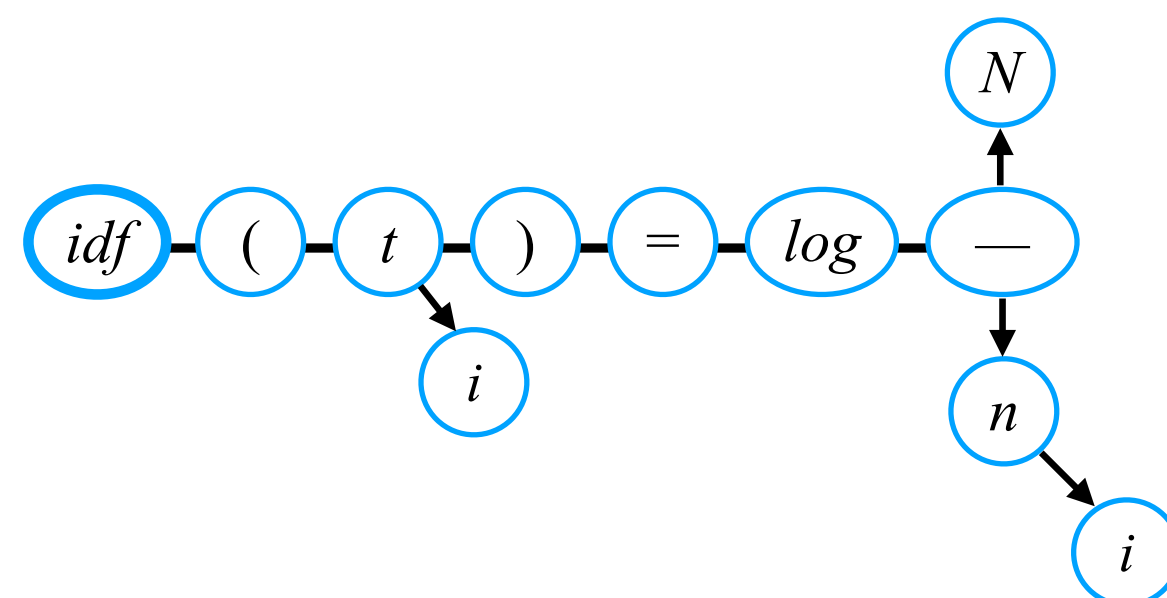
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Symbol Layout Tree (SLT) for Equation (1)

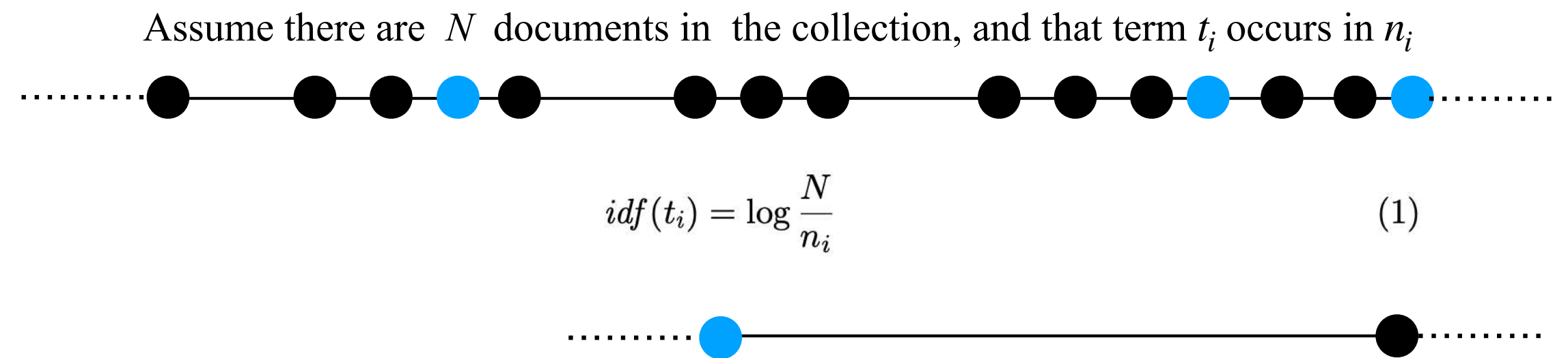


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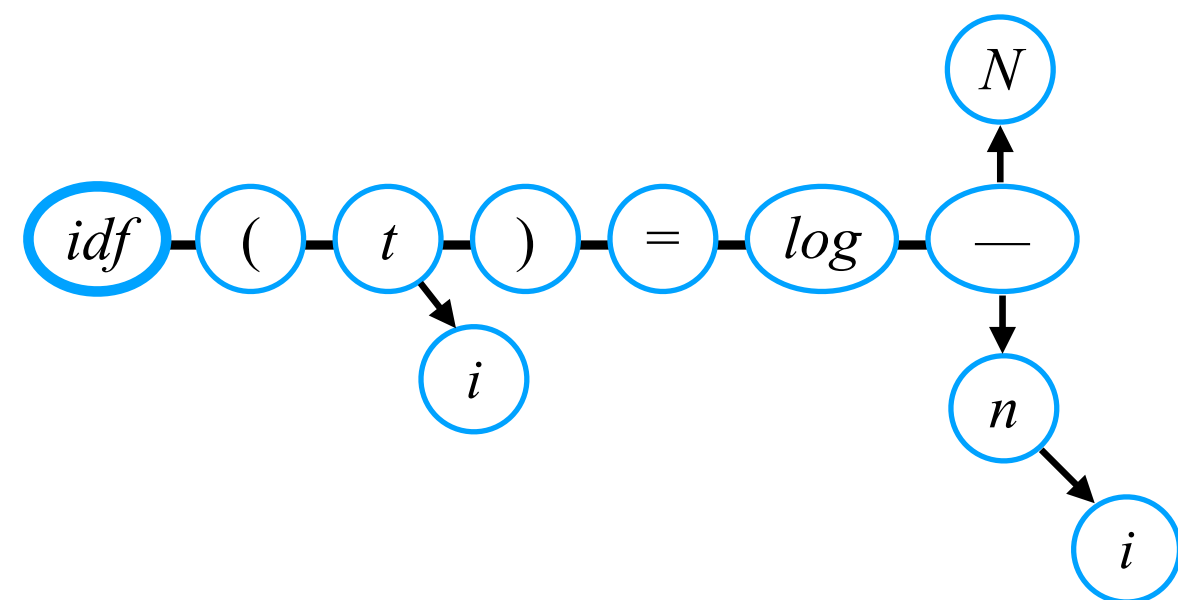
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Symbol Layout Trees

Represent formula **appearance**

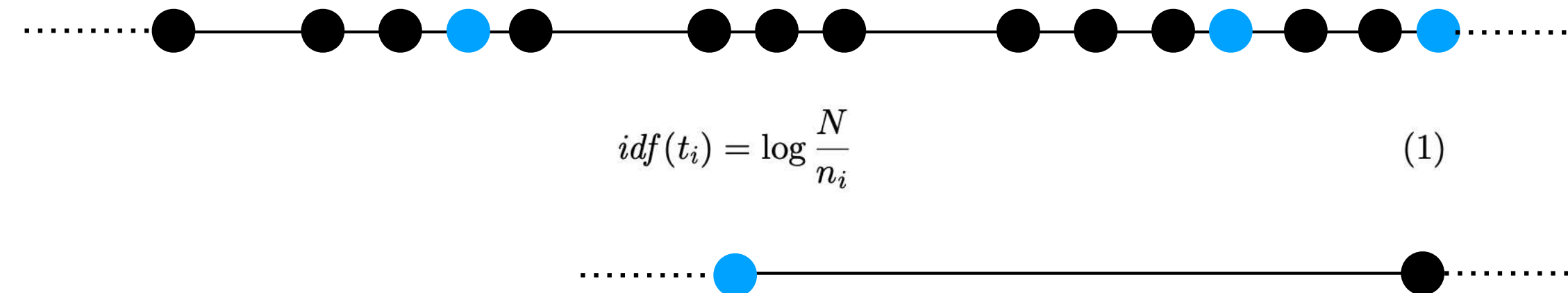
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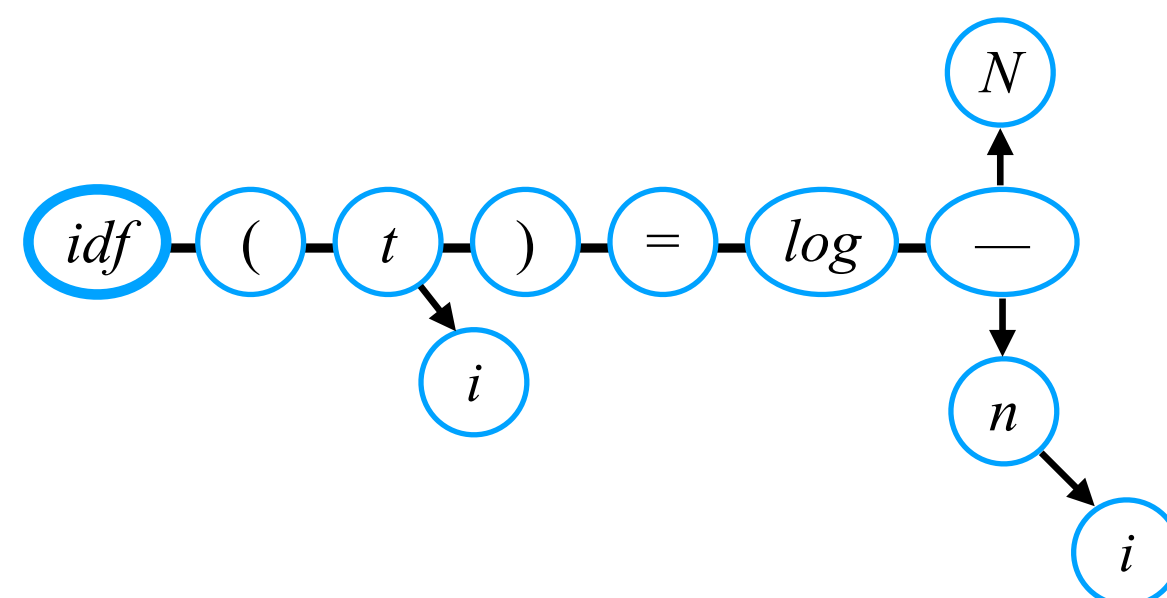
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Symbols on writing lines;

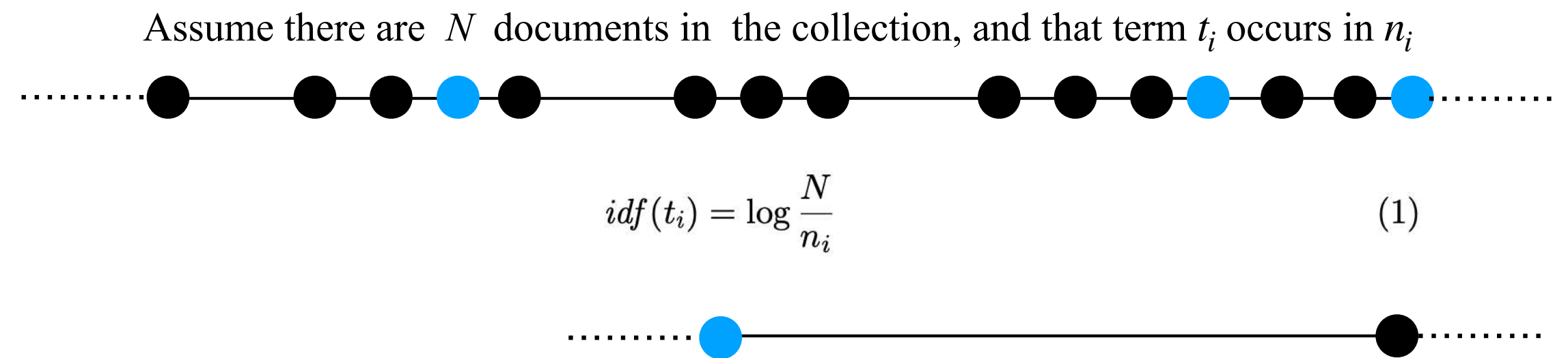
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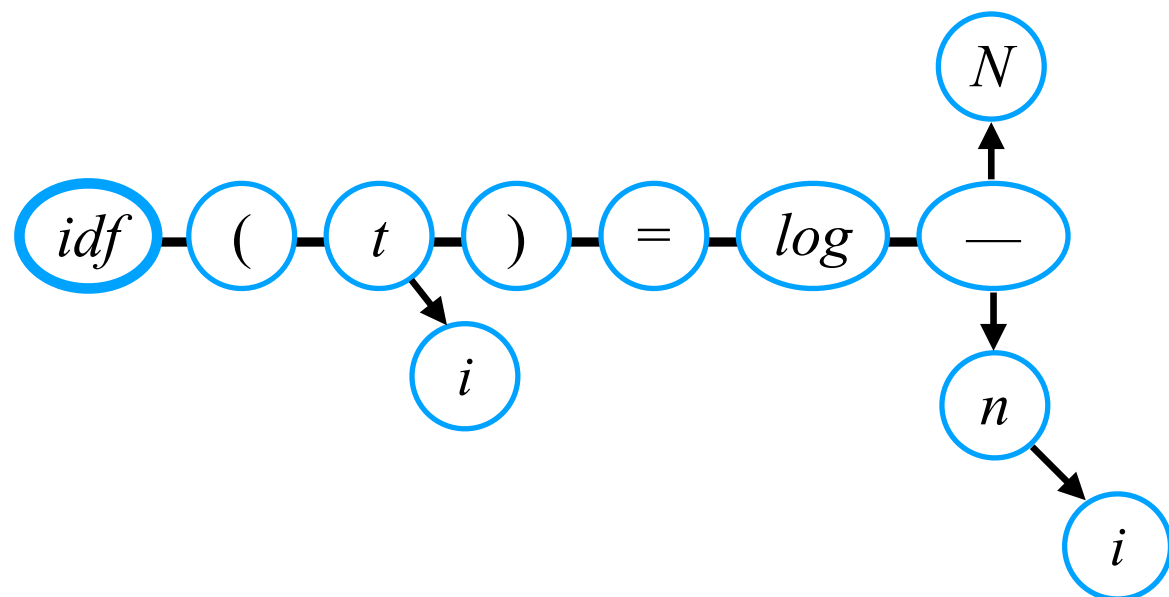
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Example 2.7: idf formula in $\text{\LaTeX}$ and Python code

$\text{\LaTeX}$: Symbol Layout Tree representation

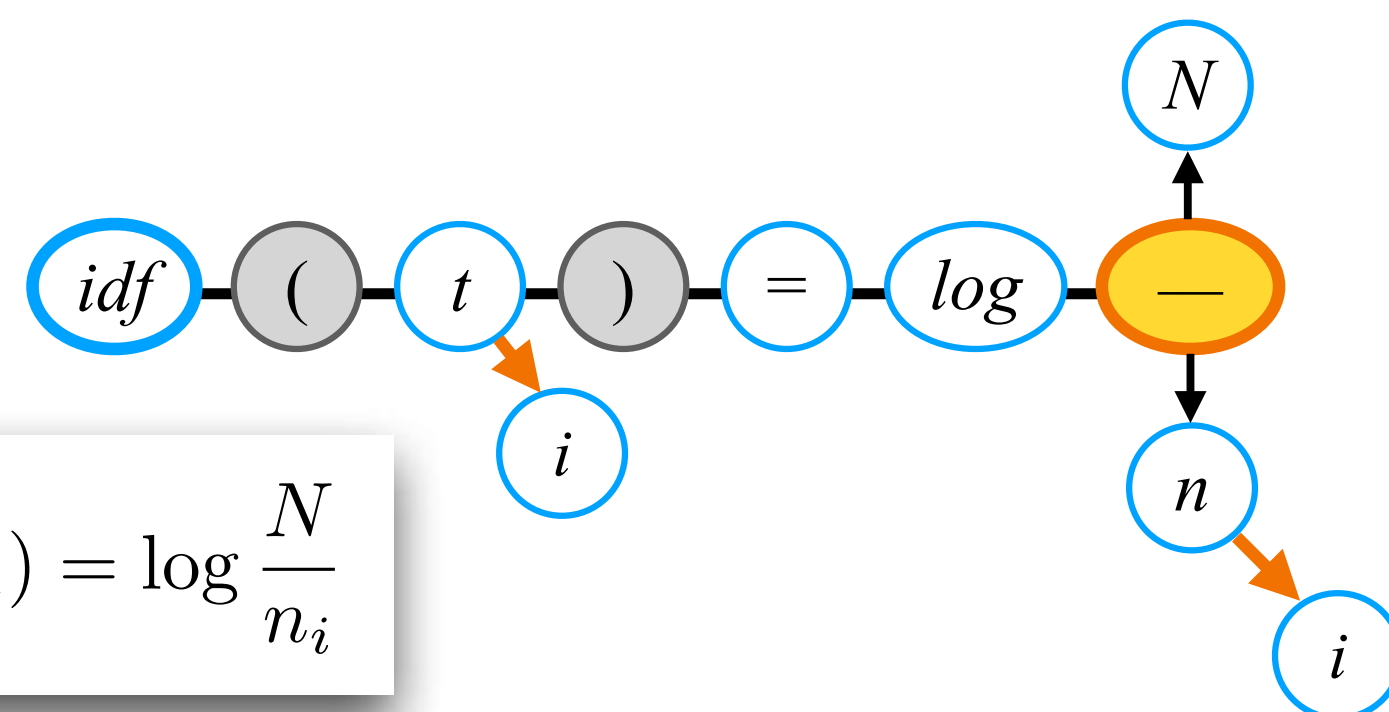
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Formulas as Operator Trees (OPTs)

Operation Hierarchy = Math operator syntax

Example 2.6: Translating a Symbol Layout Tree to an Operator Tree

Symbol Layout Tree (SLT)



Operator Tree

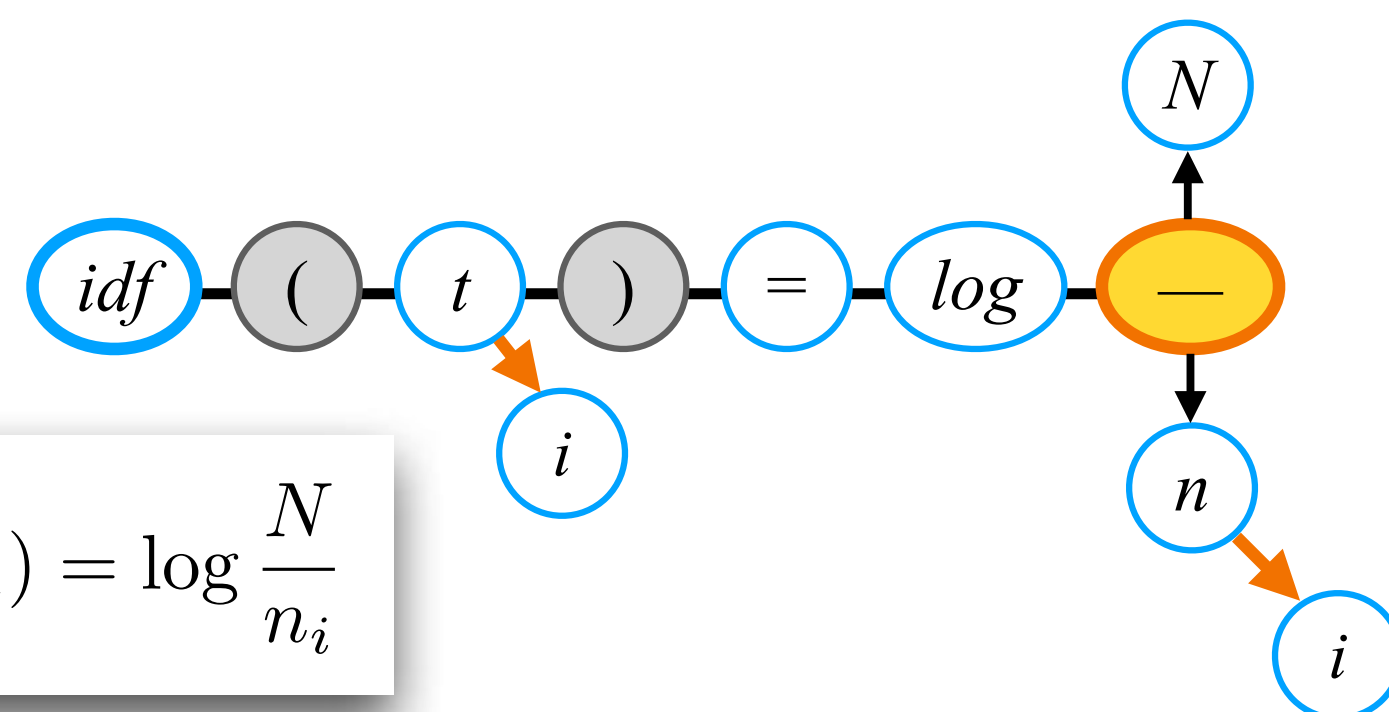
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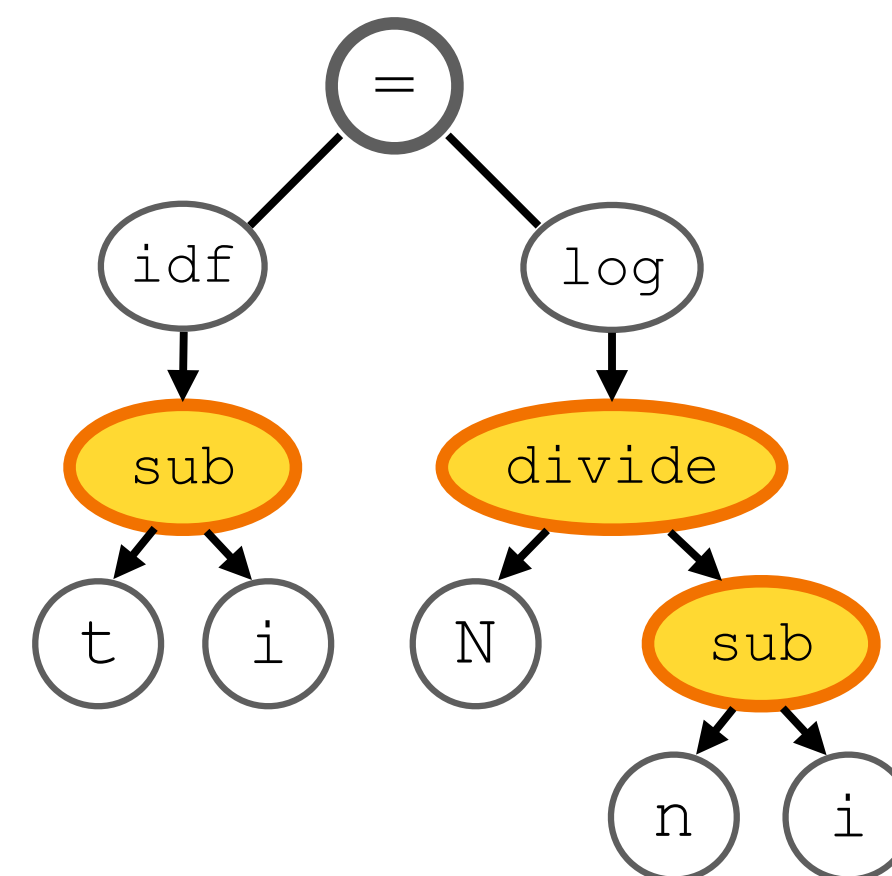
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Operator Tree (OPT)



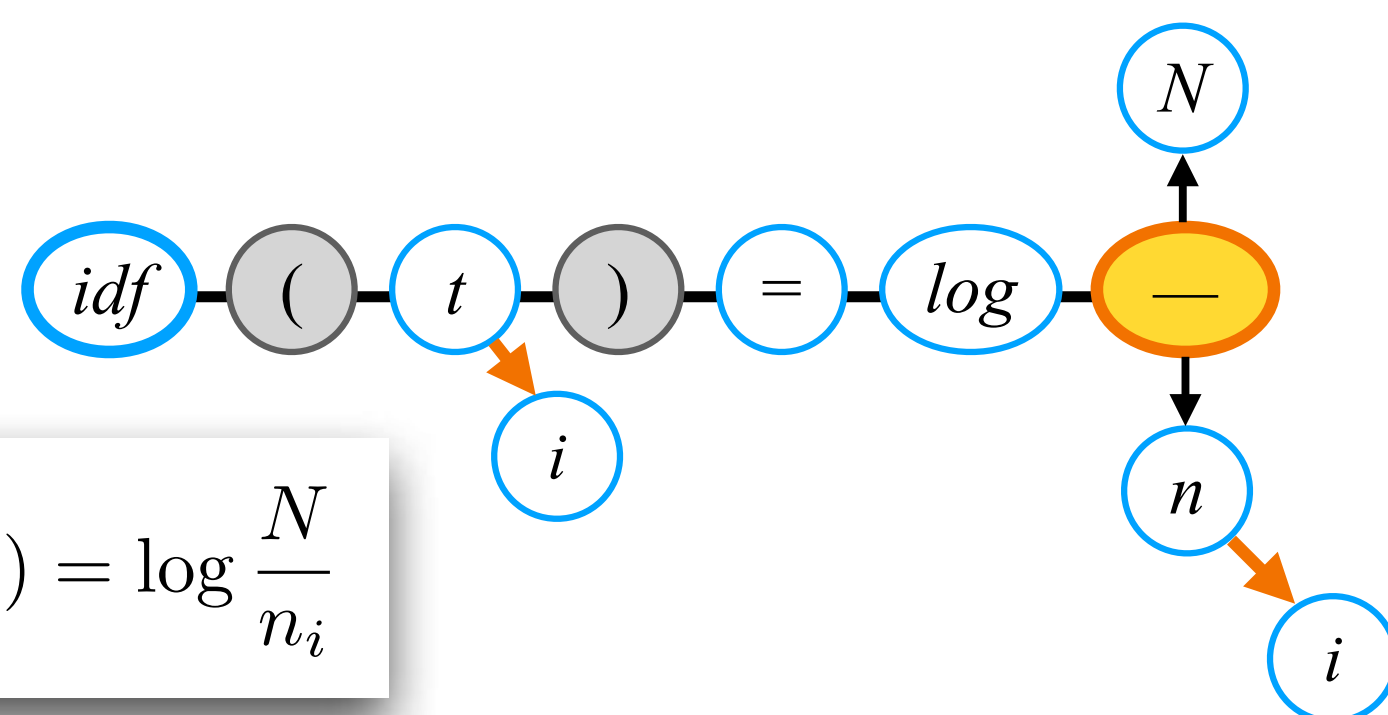
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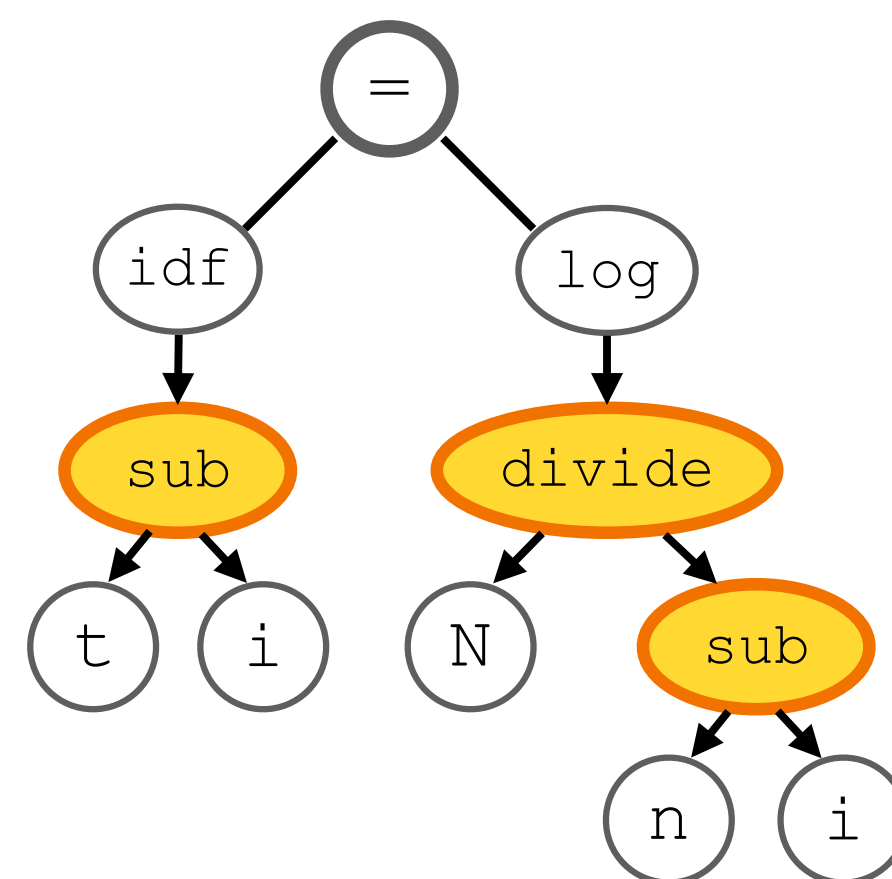
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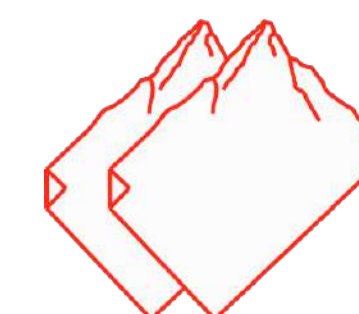
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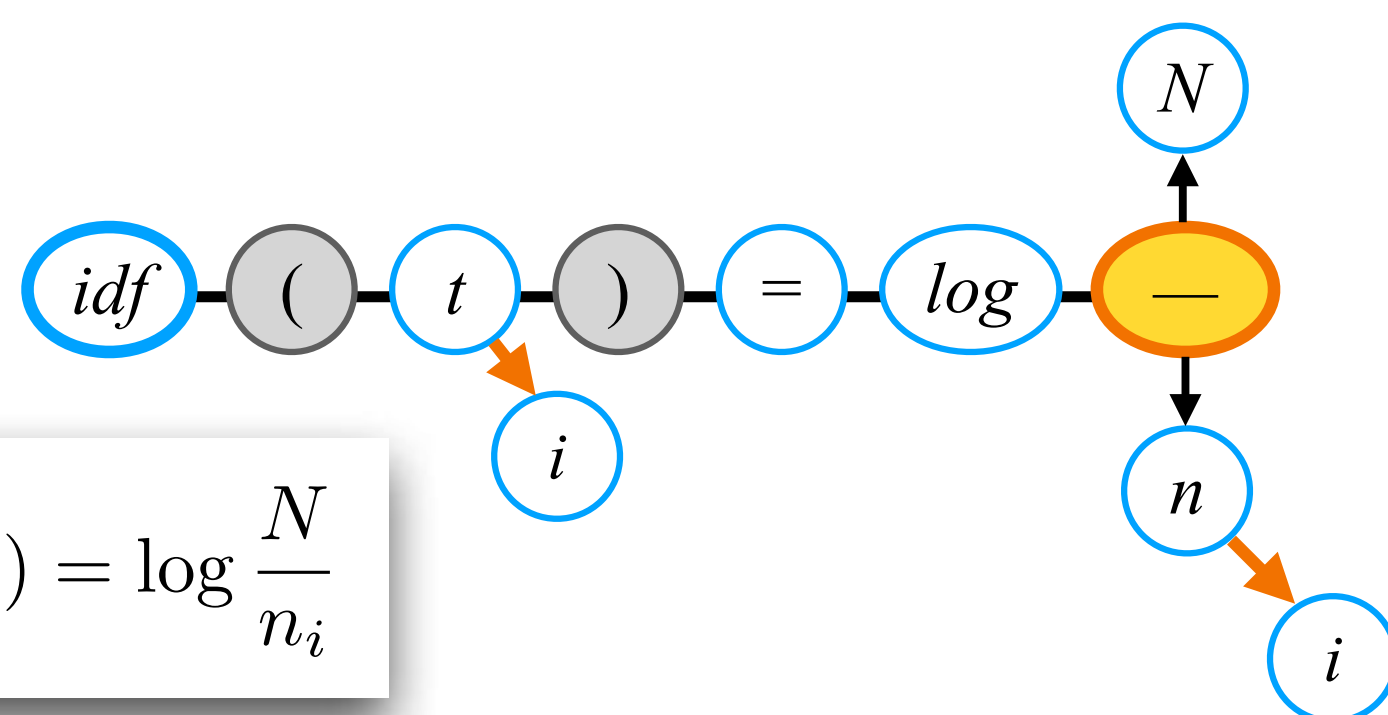
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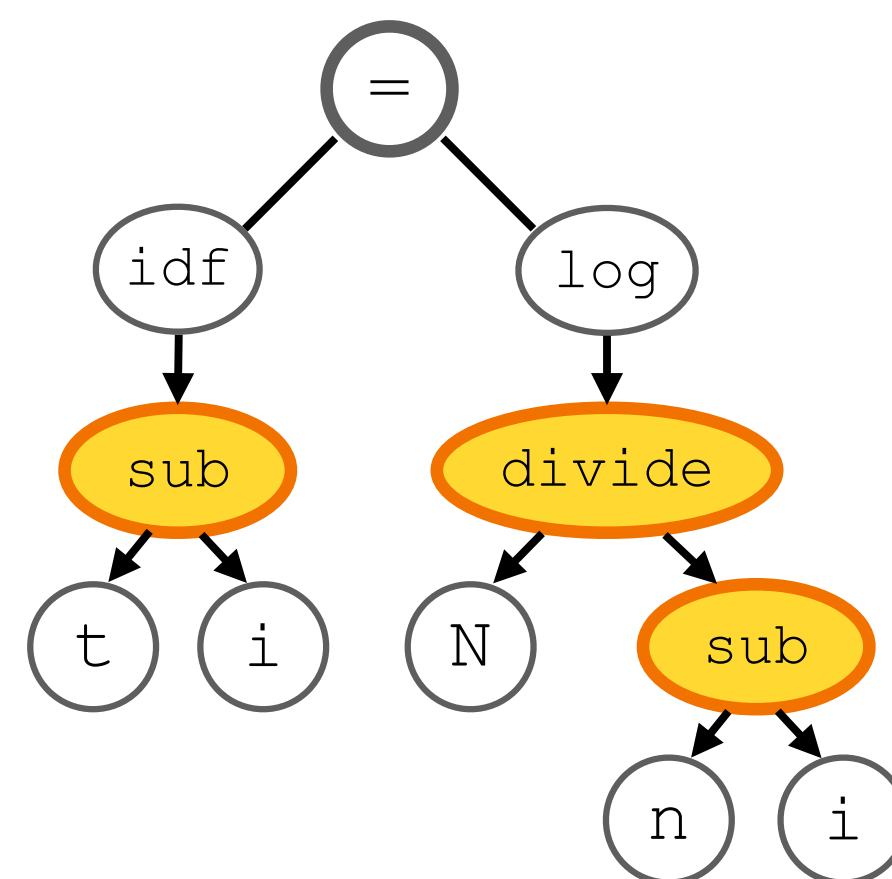
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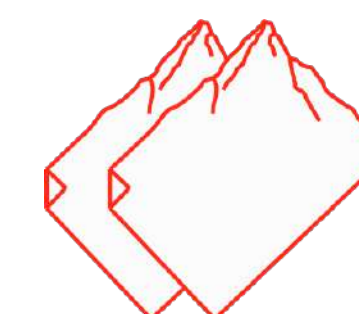
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Last operation at root

Operands are children of a node

Evaluation: **bottom-up**



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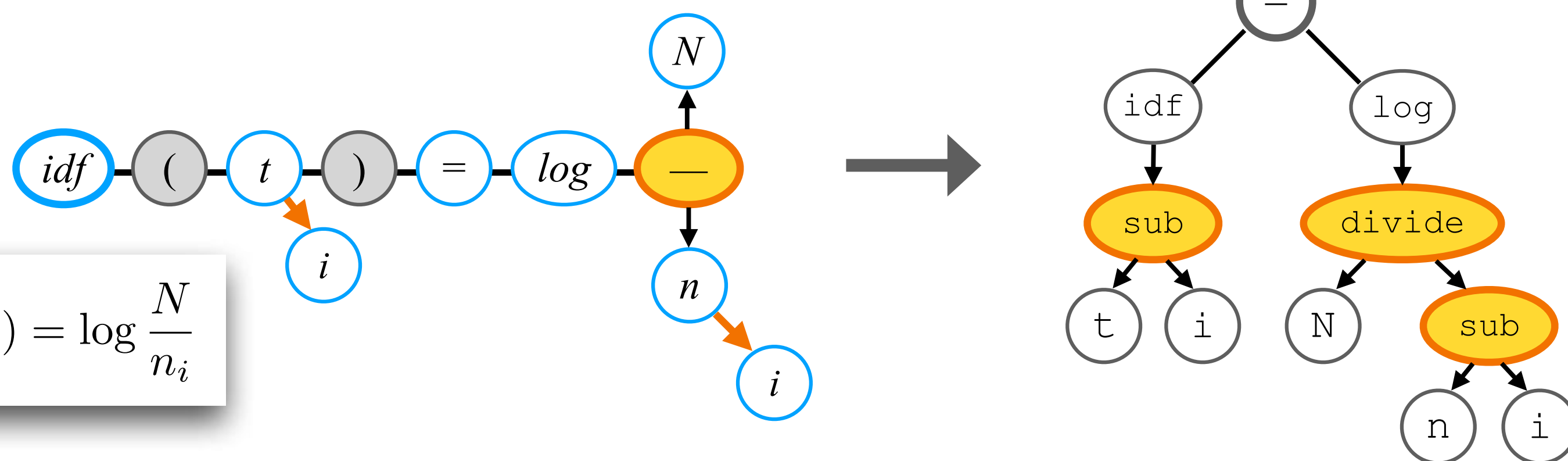
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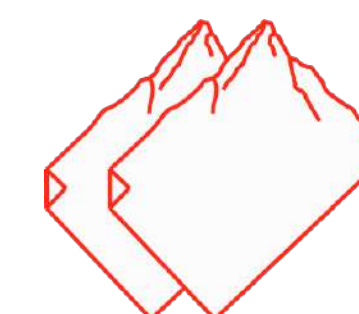
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Syntactic **NOT Semantic**

Semantics needs more information



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i.e., real-world document collections

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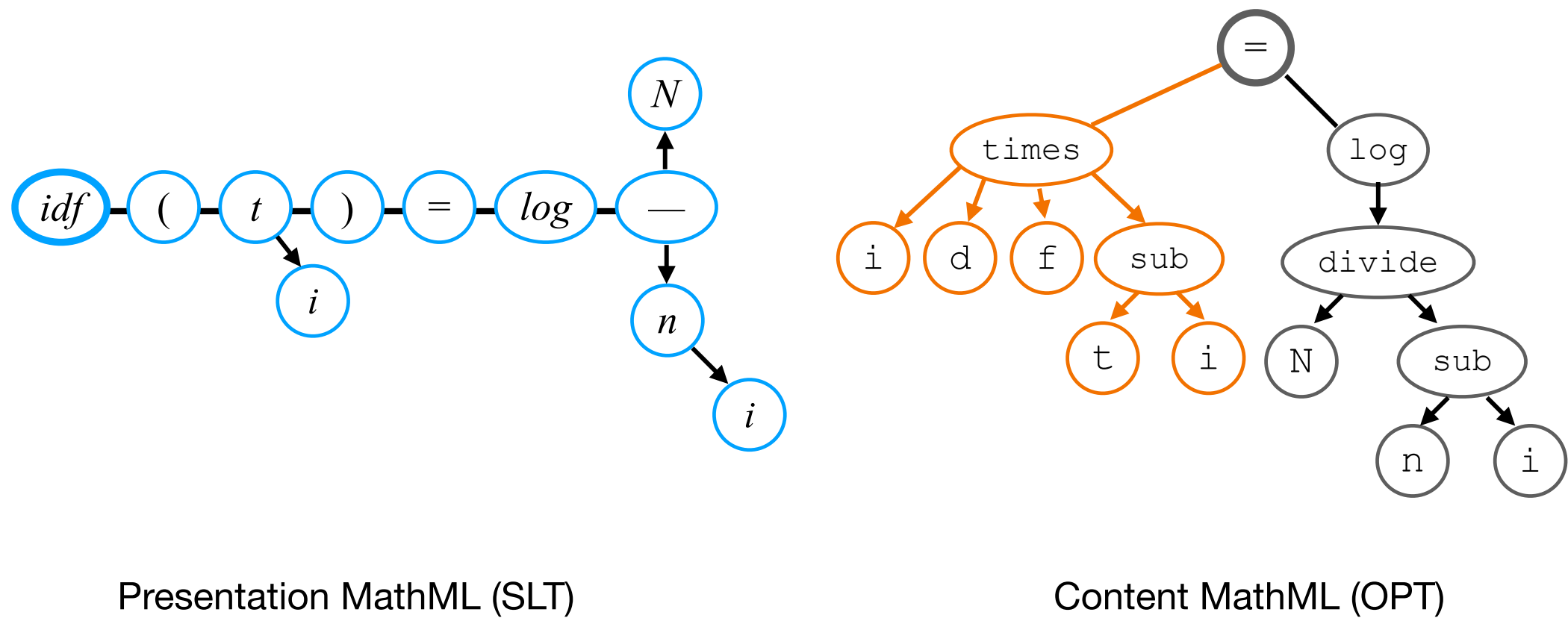
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- Operations, variables, and constants can be defined multiple ways
- Operation argument structure varies by context
- Many formulas presented ambiguously, or incompletely in docs

LaTeXML SLT → OPT Translation

Example 2.8: MathML generated from $\text{\LaTeX}$ using LaTeXML

idf is undefined in LaTeXML^a and so i , d , and f are treated as variables.



Presentation MathML (SLT)

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  <mi>d</mi>
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  <mo stretchy="false">(</mo>
  <msub>
    <mi>t</mi>
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  <mo>=</mo>
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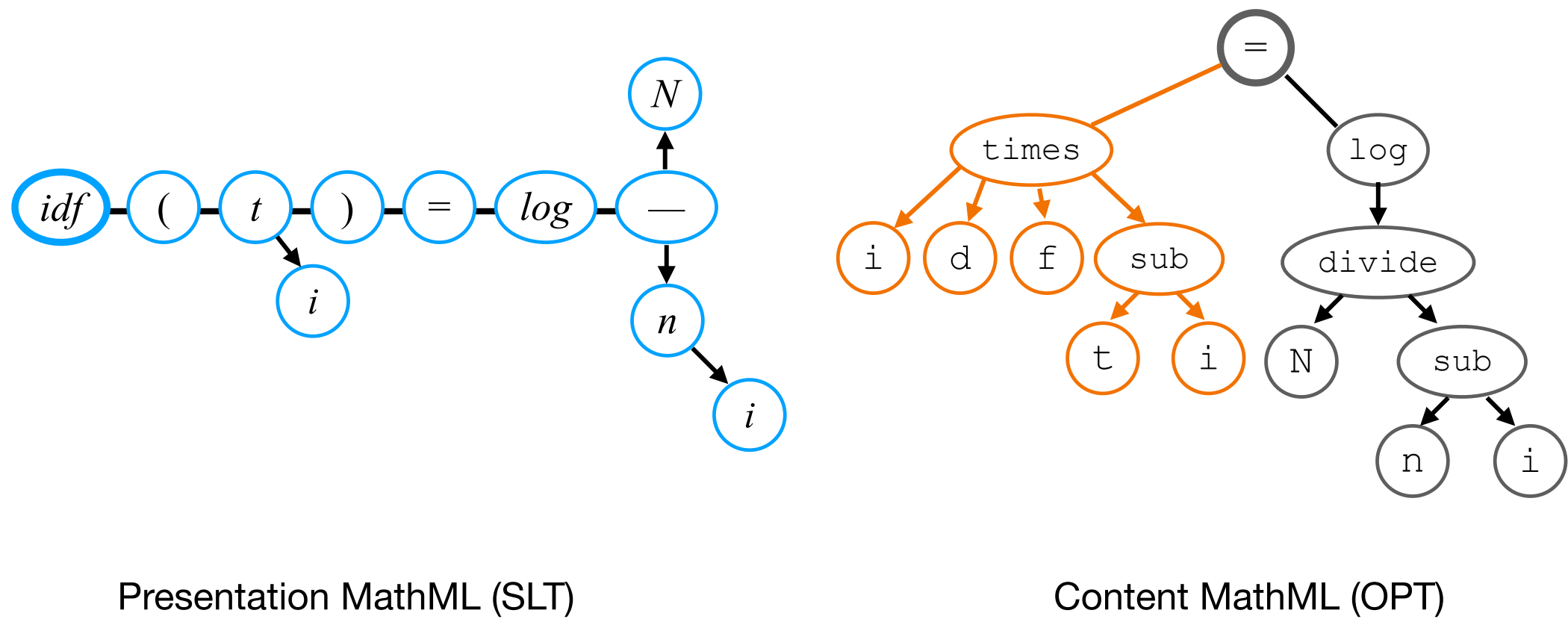
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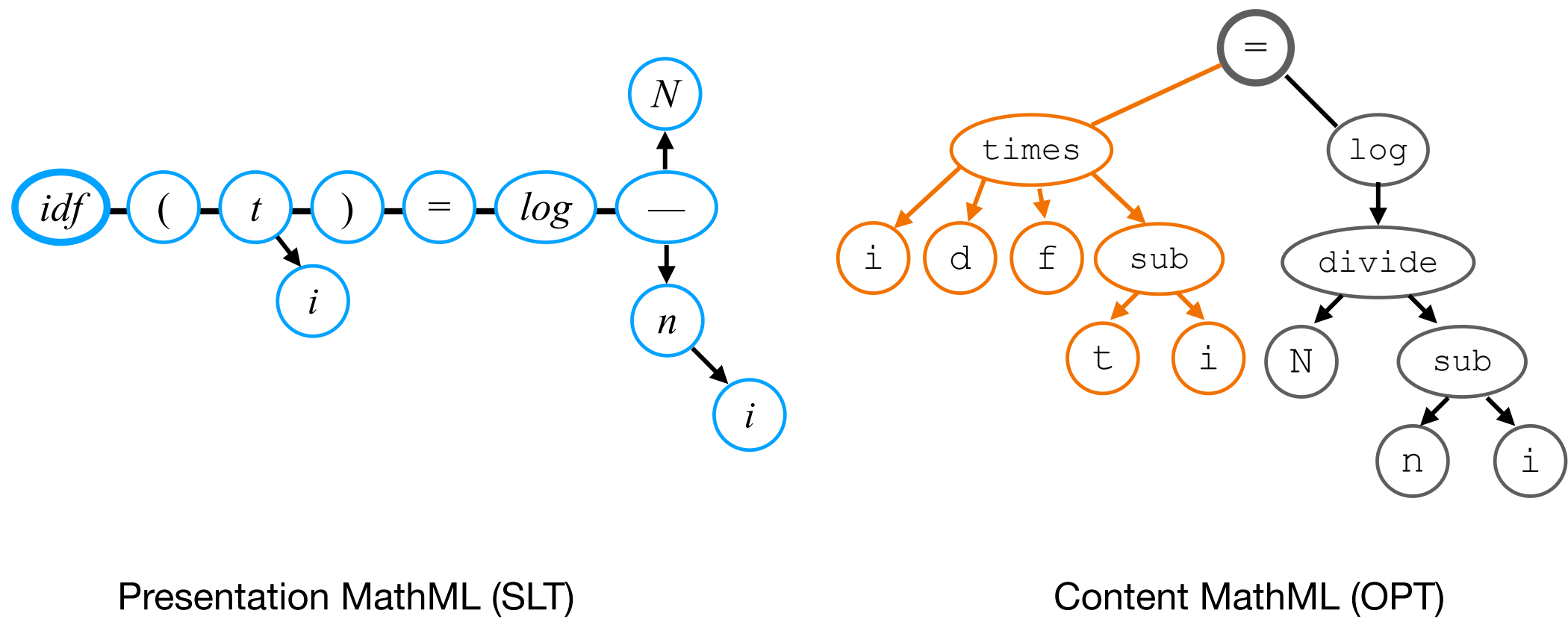
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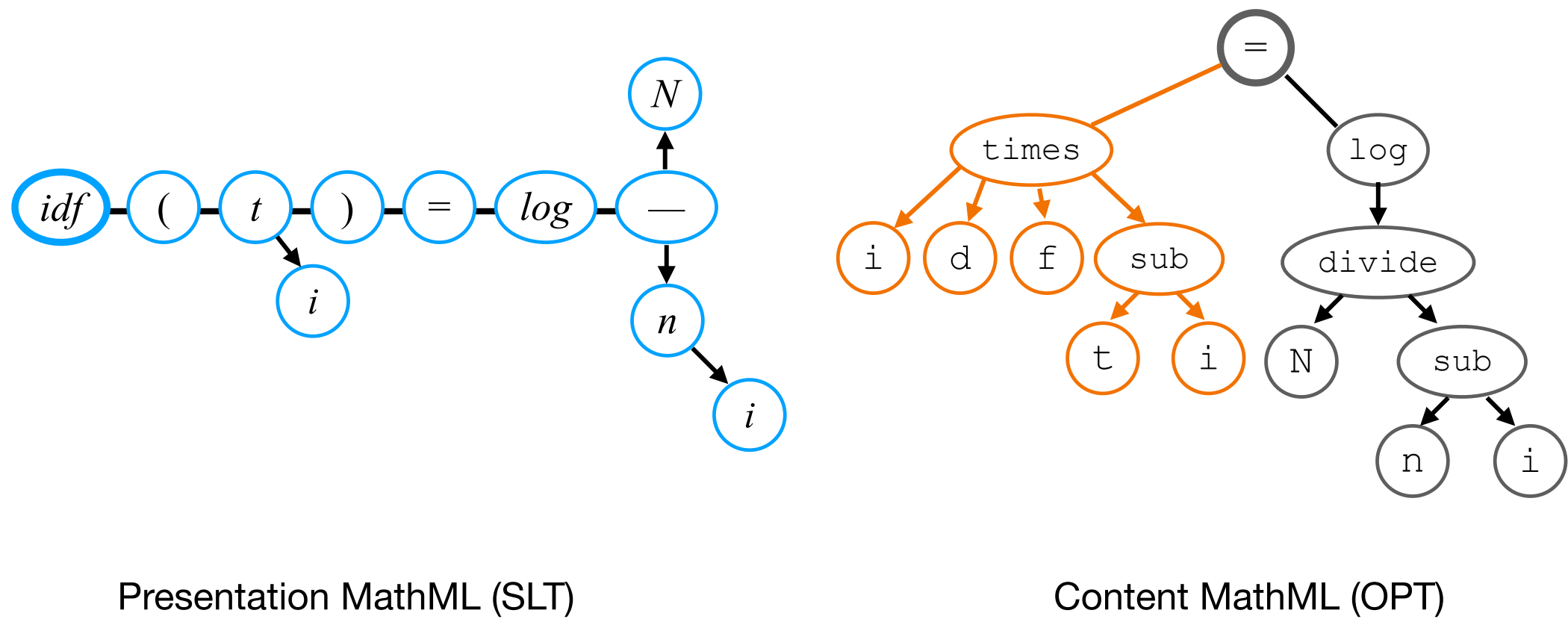
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Formula Semantics from OPT + Interpretation Context

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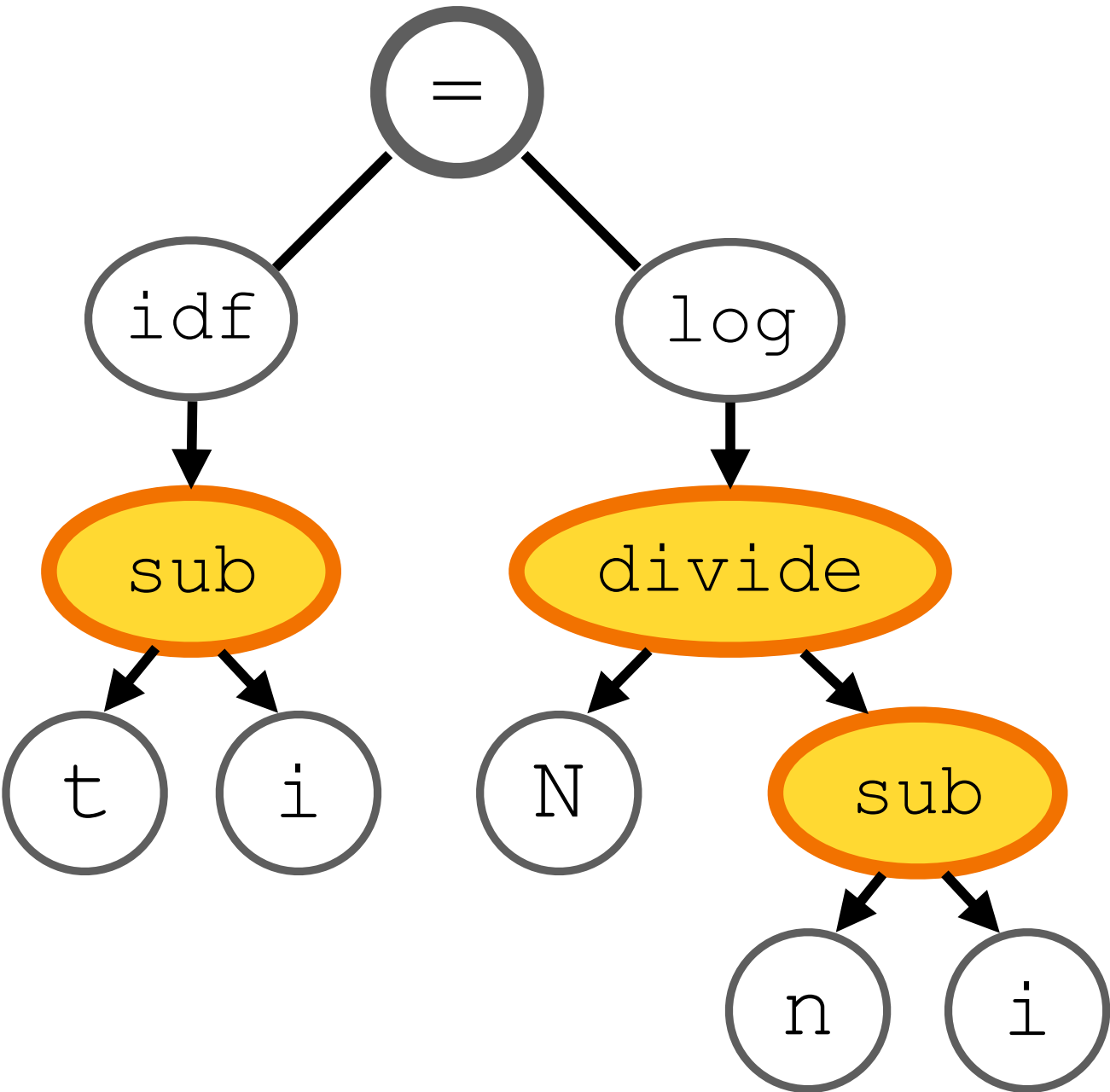
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Python
Operator Tree Code, v1

Python
Operator Tree Code, v2

Python Output (for v1)

Operator Tree (OPT)



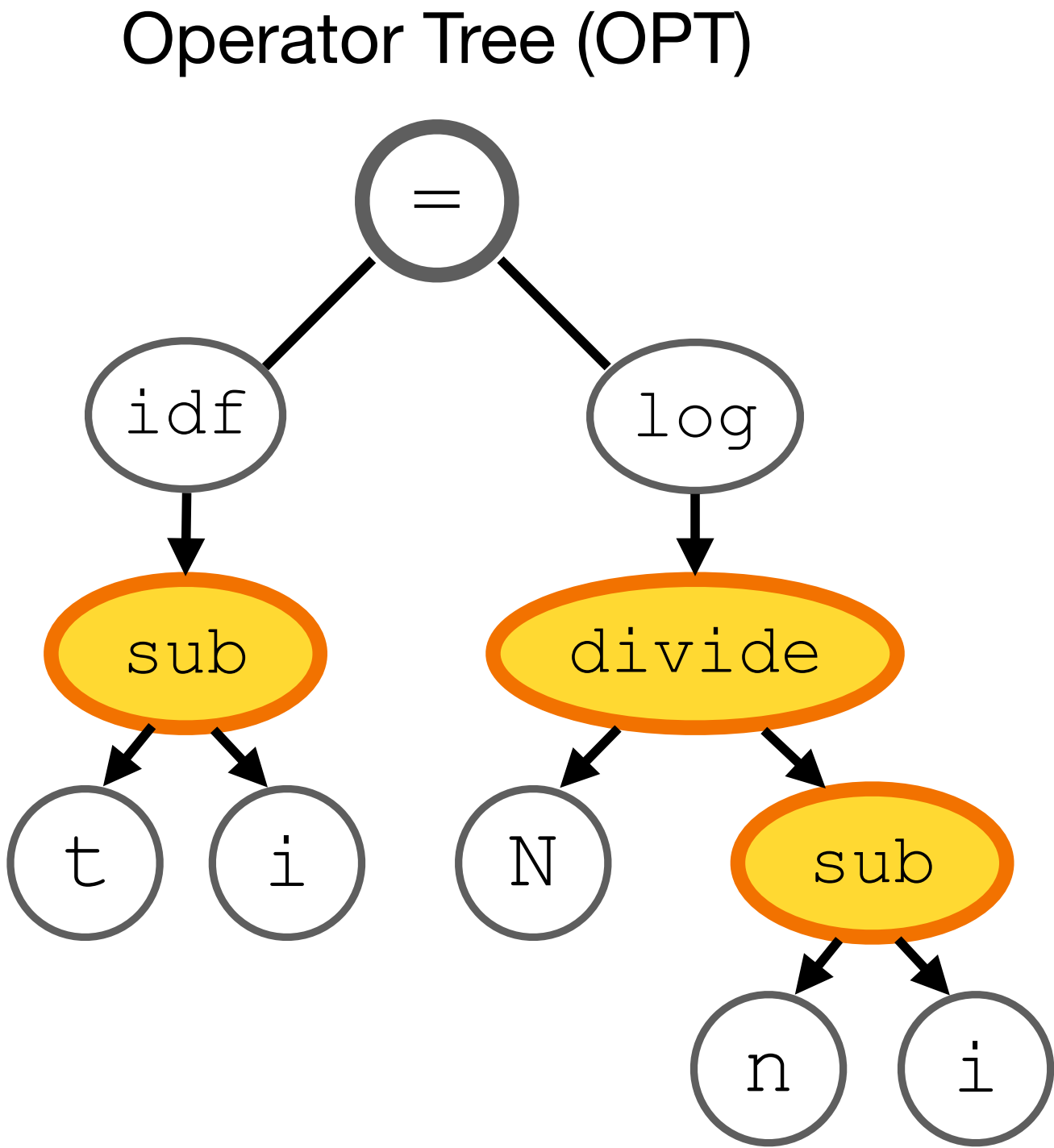
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- Interpretation Context**
- 1. = is assignment; *subscript* is array access
 - 2. Select known ops from Python functions
 - 3. Select var types & precision



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Python
Operator Tree Code, v2

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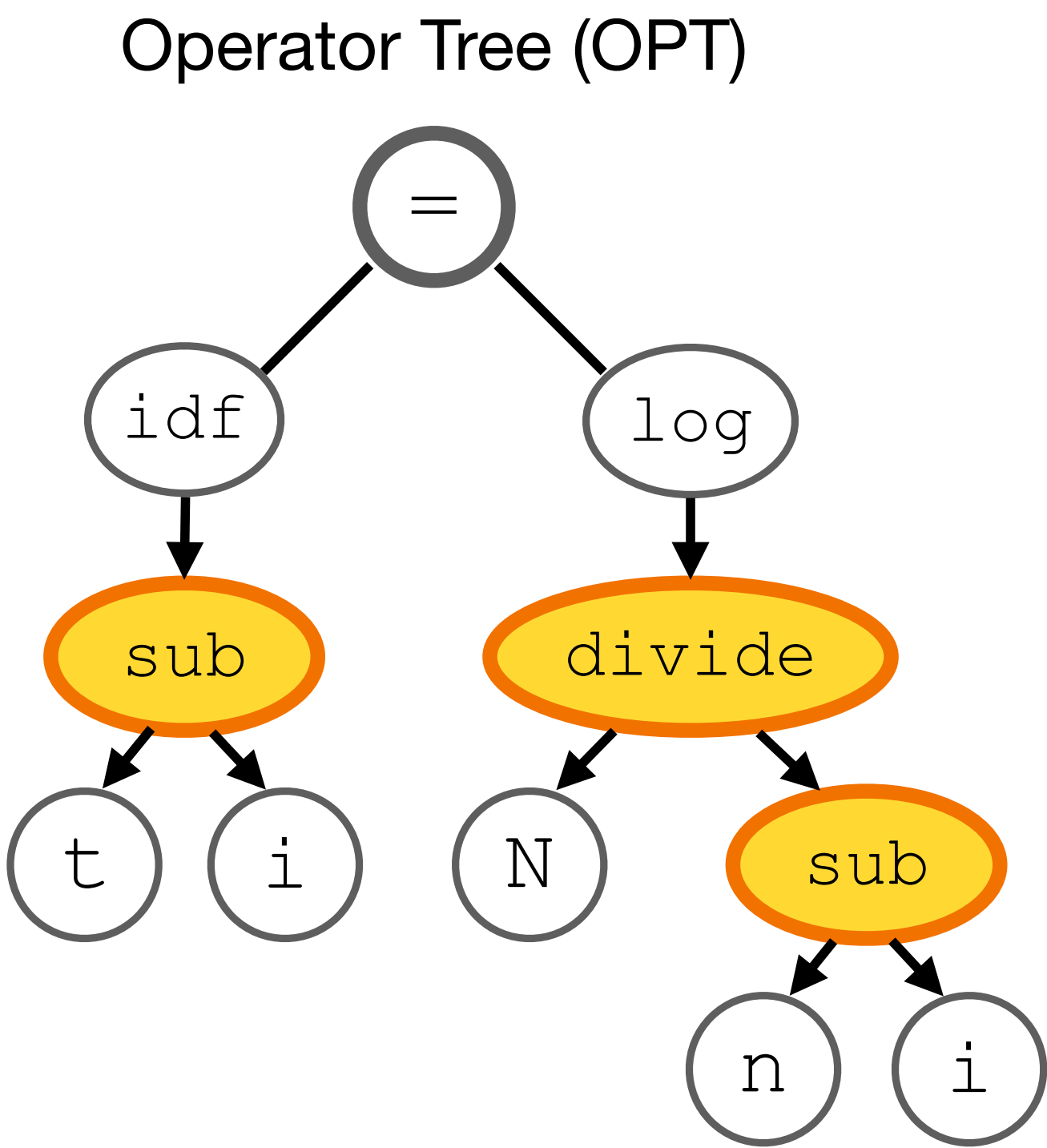
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$$idf(t_i) = \log \frac{N}{n_i}$$

- Interpretation Context**
- 1. = is assignment; *subscript* is array access
 - 2. Select known ops from Python functions
 - 3. Select var types & precision



Example 2.7: *idf* formula in L^AT_EX and Python code

Python: Two Operator Tree representations

```
import math
t_all = [ "inverse", "document", "frequency" ]
n_all = [ 2, 100, 20 ]
D = 100
def idf(i, t, n, N):
    idf_weight = math.log( N / n[i] )
    return( t[i], idf_weight )
```

Python
Operator Tree Code, v2

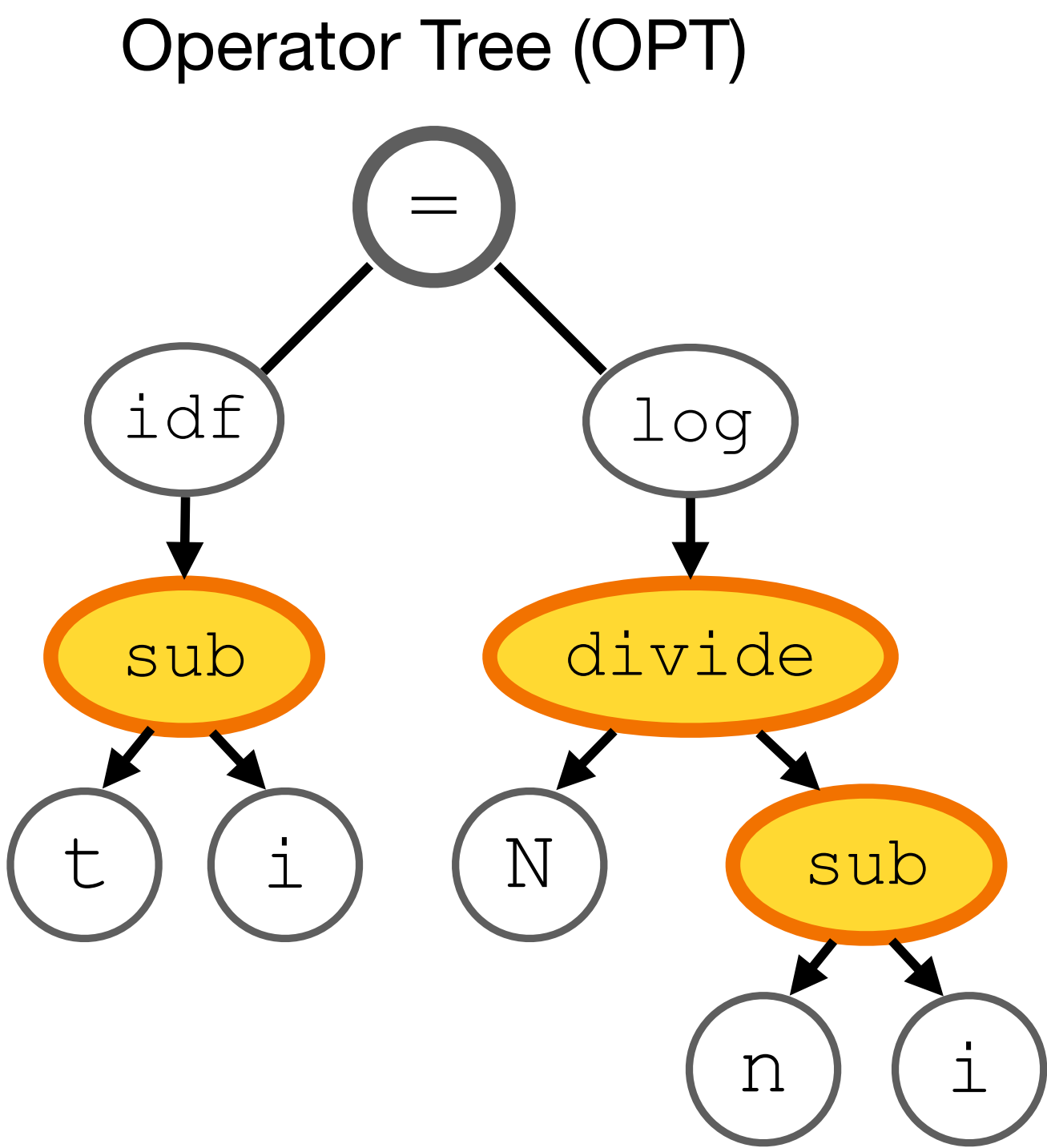
Python Output (for v1)

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```

```
# Prefix form: ops before args to match OPT RHS
def divide(a, b): return a / b
def sub(a, b): return a[b]
def idf_prefix(i, t, n, N):
    idf_weight = math.log( divide( N, sub(n, i)) )
    return( sub(t, i), idf_weight )
```

Python Output (for v1)

Formula Semantics from OPT + Interpretation Context

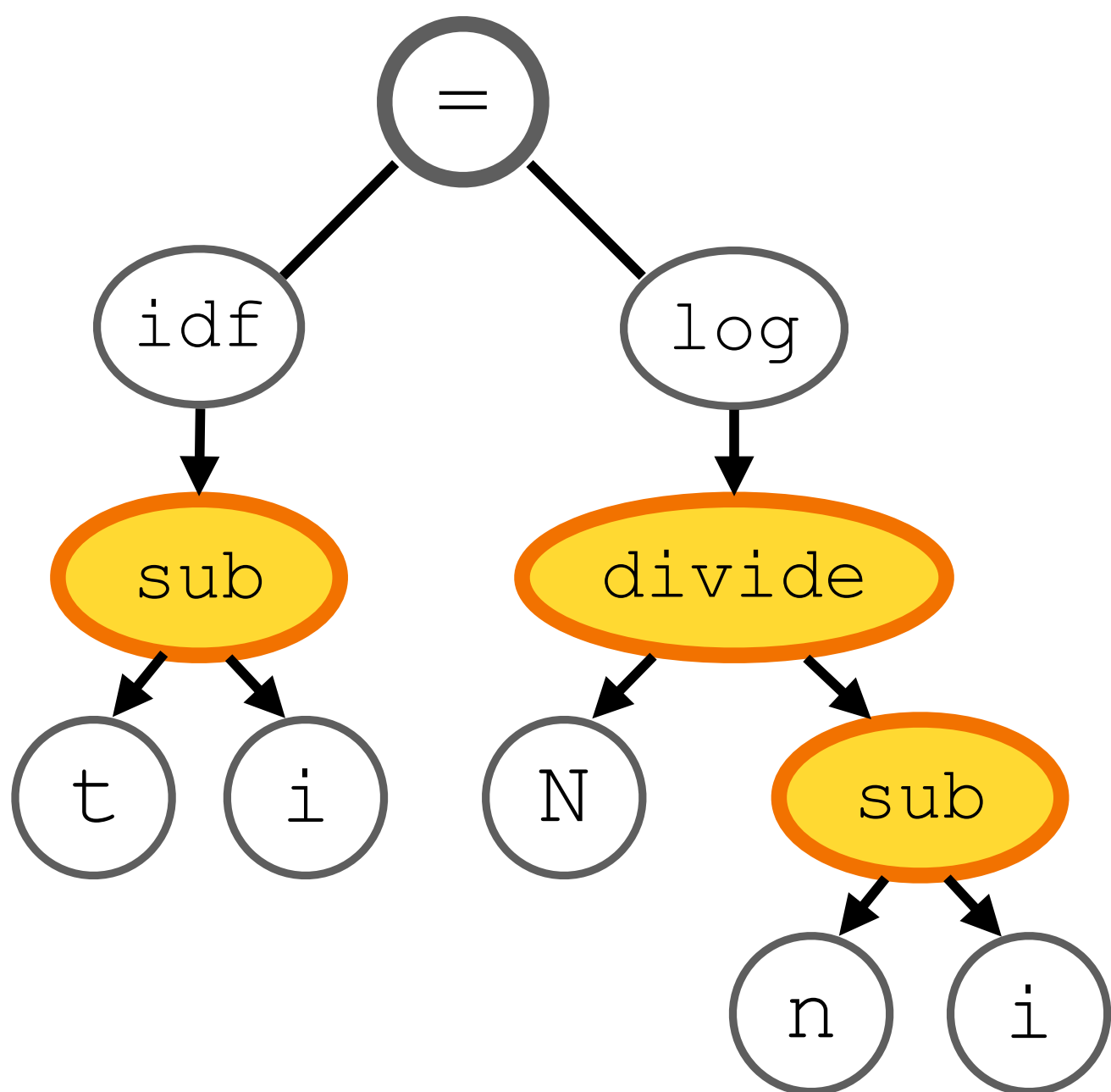
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Operator Tree (OPT)



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```

```
for i in range(len(t_all)):
    print(i, idf(i, t_all, n_all, D))
```

OUTPUT: 0 ('inverse', 3.912023005428146)
1 ('document', 0.0)
2 ('frequency', 1.6094379124341003)

Retrieving Formulas & (Multimodal) Math-Aware Search

Math-aware search (ad-hoc retrieval)

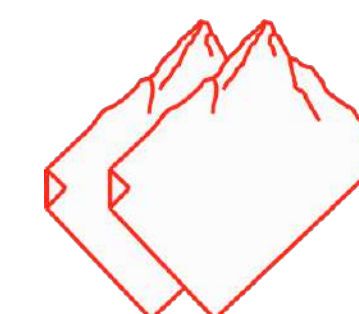
Query	Result
I have the sum $\sum_{k=0}^n \binom{n}{k} z^k$ know the result is $n^2 - 1$ but I don't know how you get there. How does one even begin to simplify a sum like this that has binomial coefficients.	1 ... which can be obtained by manipulating the second derivative of $\sum_{k=0}^n \binom{n}{k} z^k$ and let $z = p/(1 - p) \dots$ 2 Yes, it is in fact possible to sum this. The answer is $\sum_{k=0}^n \binom{n}{k} \binom{m}{k} = \binom{m+n}{n}$ assuming that $n \leq m$. This comes from the fact that ...

Formula Search Tasks

Formula Similarity

Formula Similarity
w. Wildcards

Contextualized
Formula Search



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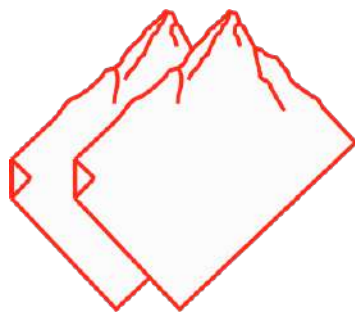
Formula Search Tasks

Formula similarity search

Query	Result				
	1	2	3	4	5
$y = \frac{a+bx}{b-x}$	$y = \frac{a+bx}{c+x}$	$y = a + bx$	$y = \frac{a-bx}{c-x}$	$y = \frac{a+bx}{x+c}$	$g(x) = \frac{x}{x-a}$

Formula Similarity
w. Wildcards

Contextualized
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Formula Search Tasks

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Formula similarity with wildcards (?w) (Aizawa *et al.*, 2013)

Query	Result	
$\frac{?f(?v + ?d) - ?f(?v)}{?d}$	1	$g'(cx) = \lim_{h \rightarrow 0} \frac{g(\mathbf{cx} + \mathbf{h}) - g(\mathbf{cx})}{\mathbf{h}}$
	...	...

Contextualized
Formula Search

Formula Search Tasks

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Contextualized formula search

Query	Result
I have the sum $\sum_{k=0}^n \binom{n}{k} k$ know the result is $n^2 - 1$ but I don't know how you get there. How does one even begin to simplify a sum like this that has binomial coefficients.	1 ... which can be obtained by manipulating the second derivative of $\sum_{k=0}^n \binom{n}{k} z^k$ and let $z = p/(1 - p)$... 2 Yes, it is in fact possible to sum this. The answer is $\sum_{k=0}^n \binom{n}{k} \binom{m}{k} = \binom{m+n}{n}$ assuming that $n \leq m$. This comes from the fact that ...

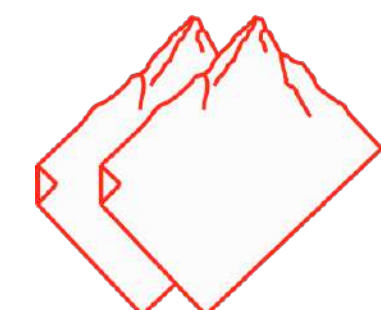
Math-Aware Search

Searching with Formulas & Text ('Multimodal')

Ad-hoc Retrieval

Math Question Answering

Word Problems



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Math-Aware Search

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Math-Aware Search

Searching with Formulas & Text (‘Multimodal’)

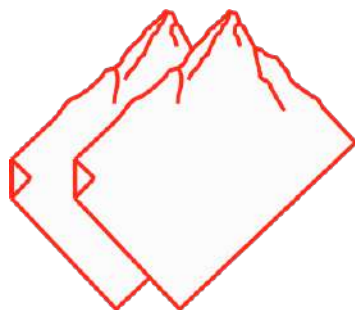
Math-aware search (ad-hoc retrieval)

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Math Question Answering (Mansouri *et al.*, 2022a)

Query	Result
What does it mean for a matrix to be Hermitian?	A matrix is Hermitian if it is equal to its transpose conjugate.

Word Problems



Math-Aware Search

Searching with Formulas & Text (‘Multimodal’)

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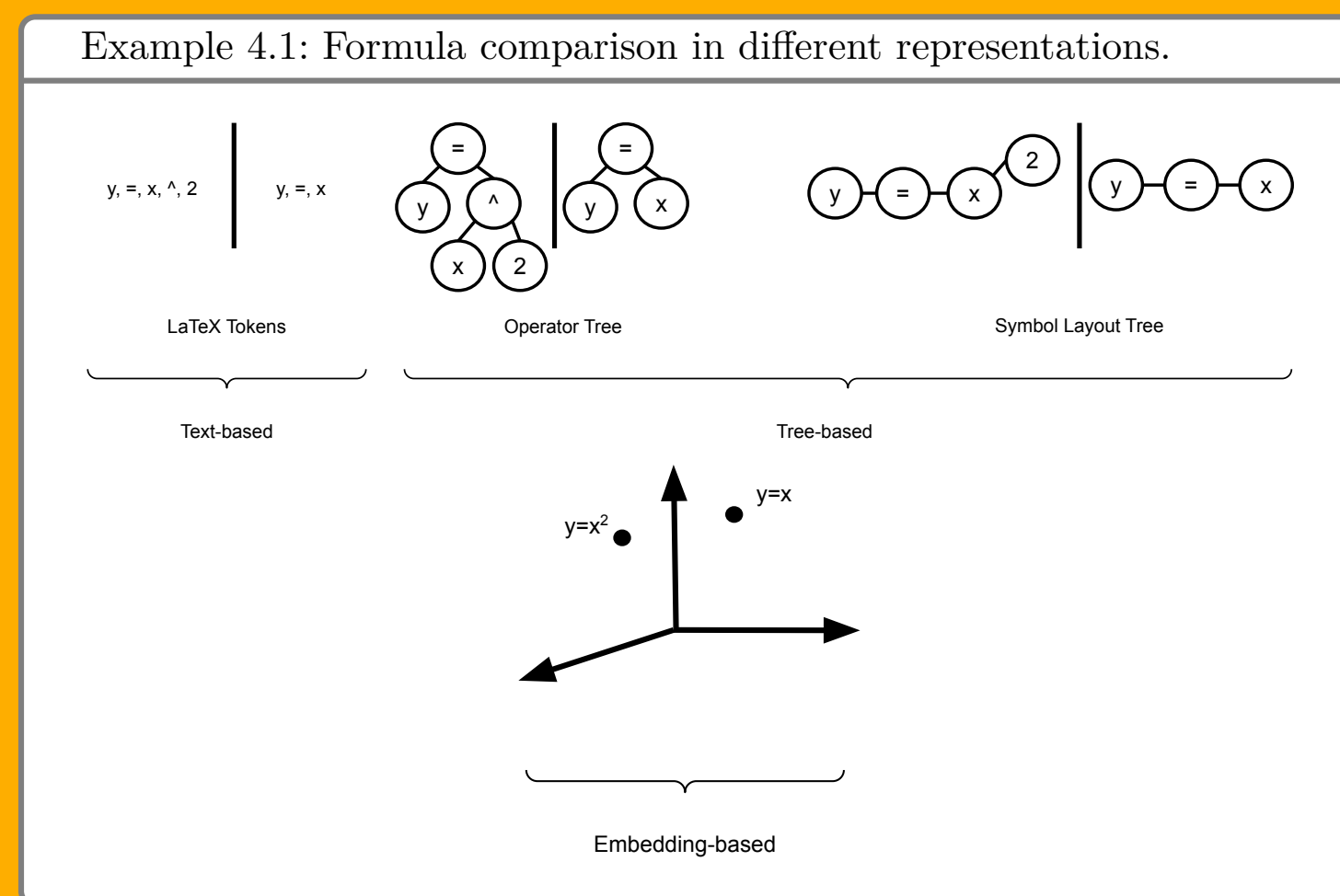
Math word problems

Query	Result	
	<i>Equation</i>	<i>Answer</i>
Sarah has 5 pens, David has 3 pens. How many pens do they have?	$x = 5 + 3$	8
Find two consecutive integers whose sum is 7.	$x + (x + 1) = 7$	3, 4

From MathQA (Amini *et al.*, 2019) & Dolphin18K (Huang *et al.*, 2016)

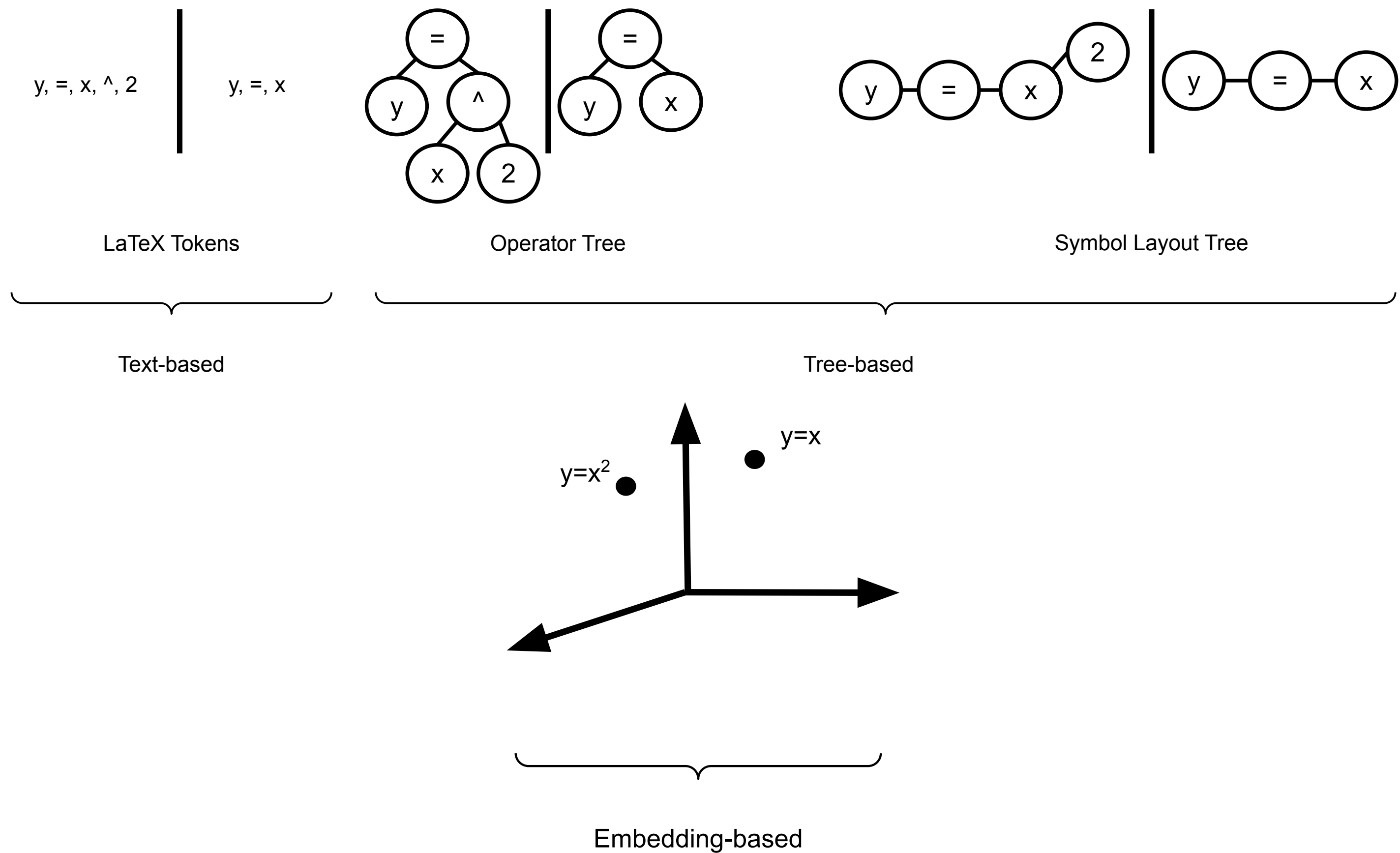


Representing Math Formulas for Retrieval



Common Representations for Formula Retrieval

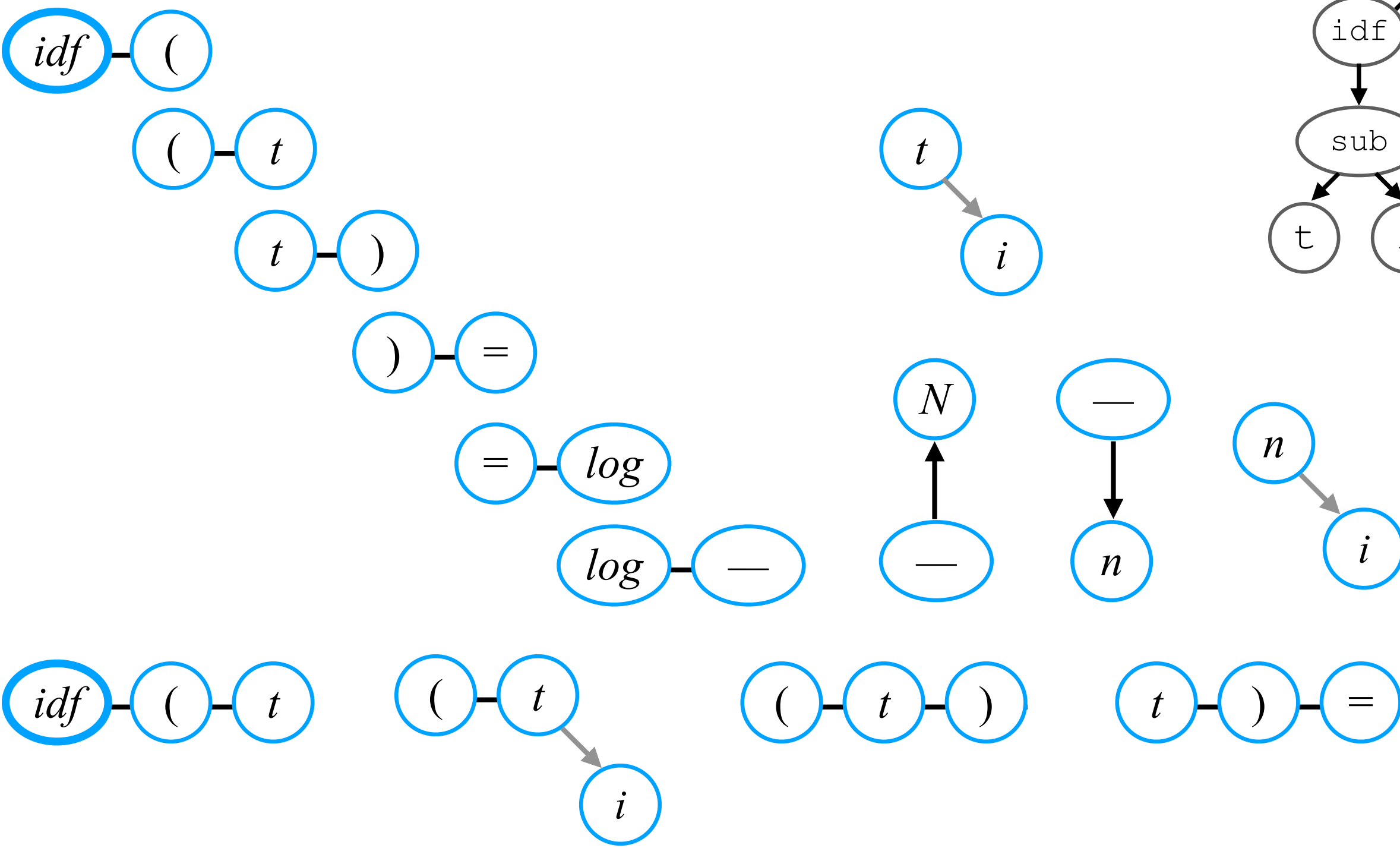
Example 4.1: Formula comparison in different representations.



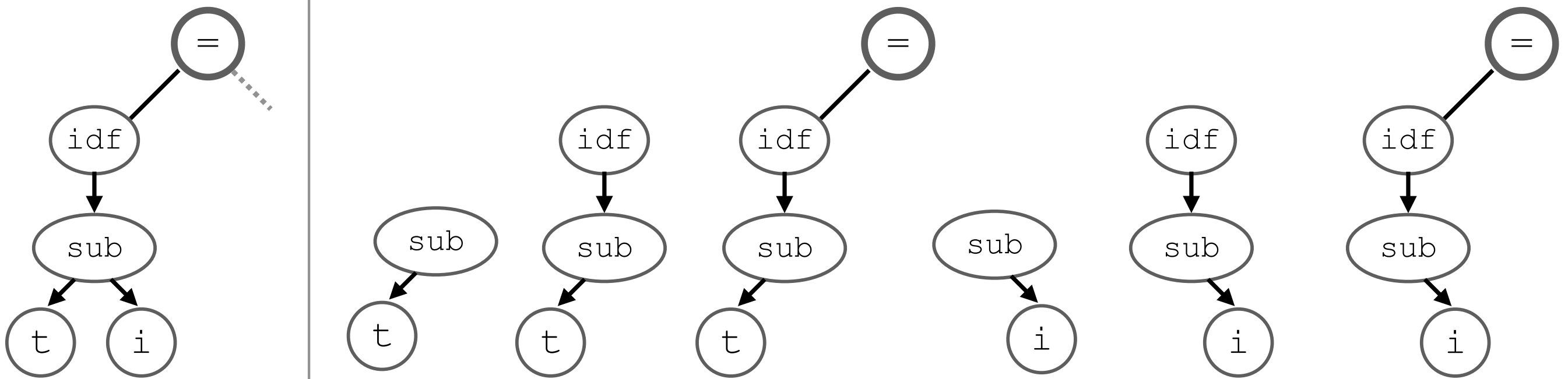
Our choice of representation(s) limit the **types of information**, and **patterns** within information types that can be searched **effectively**

Path-Based Indexing

Symbol Layout Tree Paths (directed)



Operator Tree Paths (leaf-root)

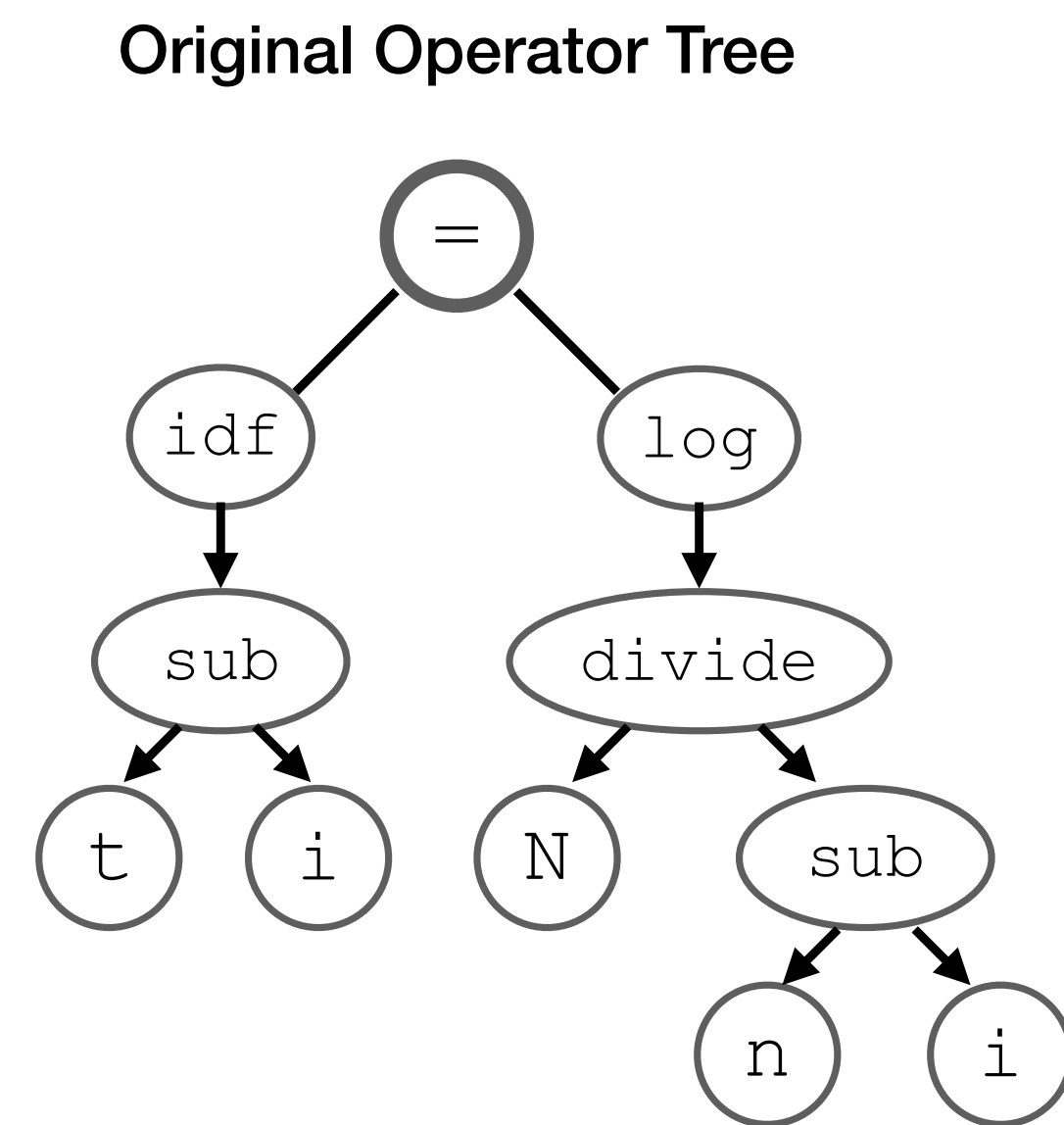


**Also used to create inputs to many embedding-based retrieval models*

Common Abstractions for Formula Search

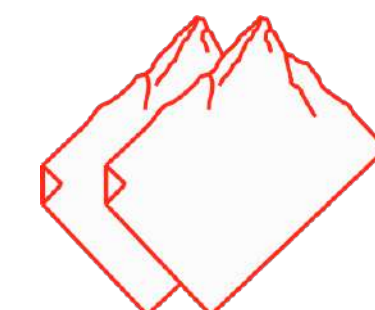
Variable Enumeration and Types

Example 2.10: OPT Variable Enumeration and Symbol Types



Variable
Enumeration

Assigning Types
Symbols/Nodes

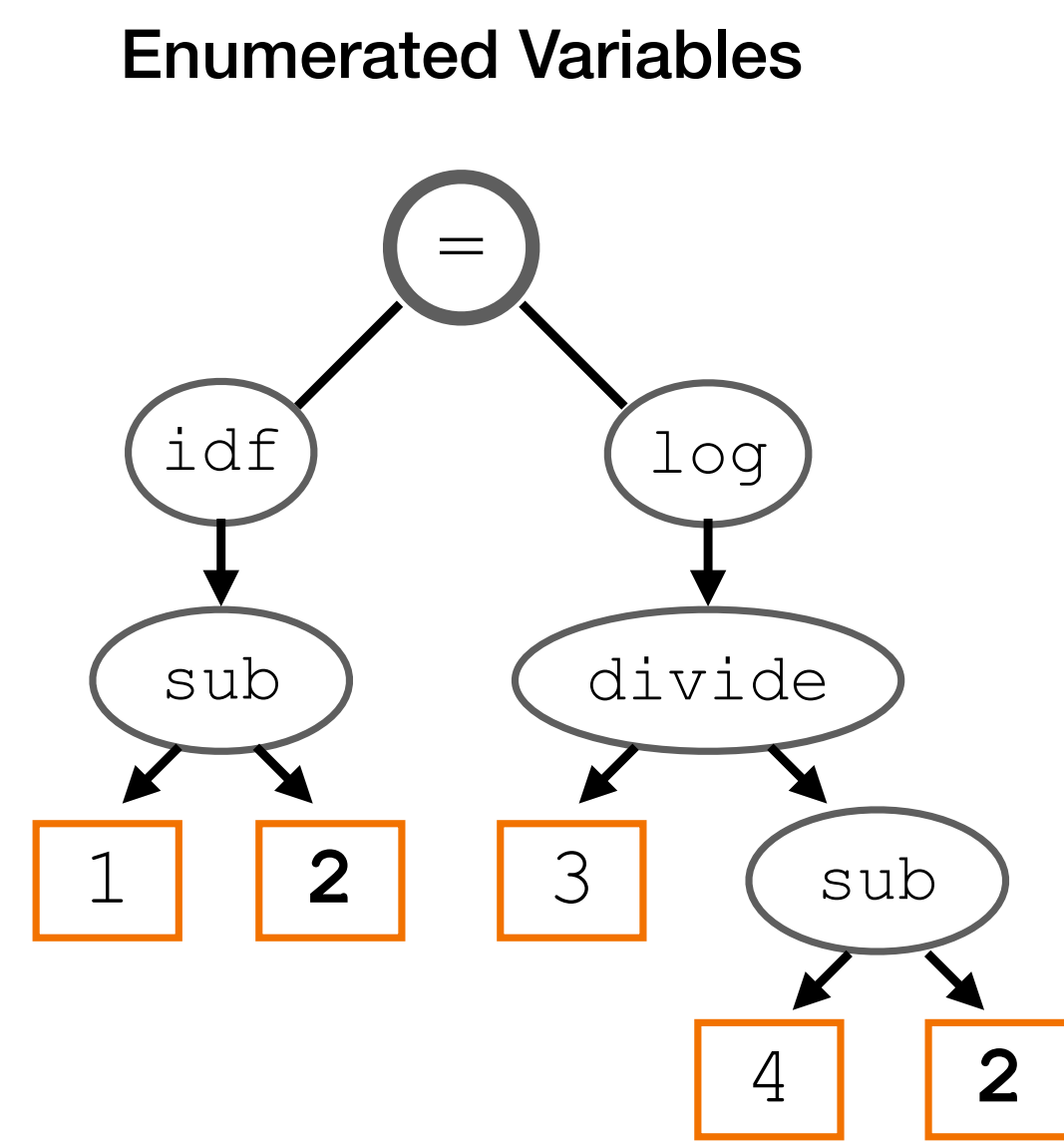
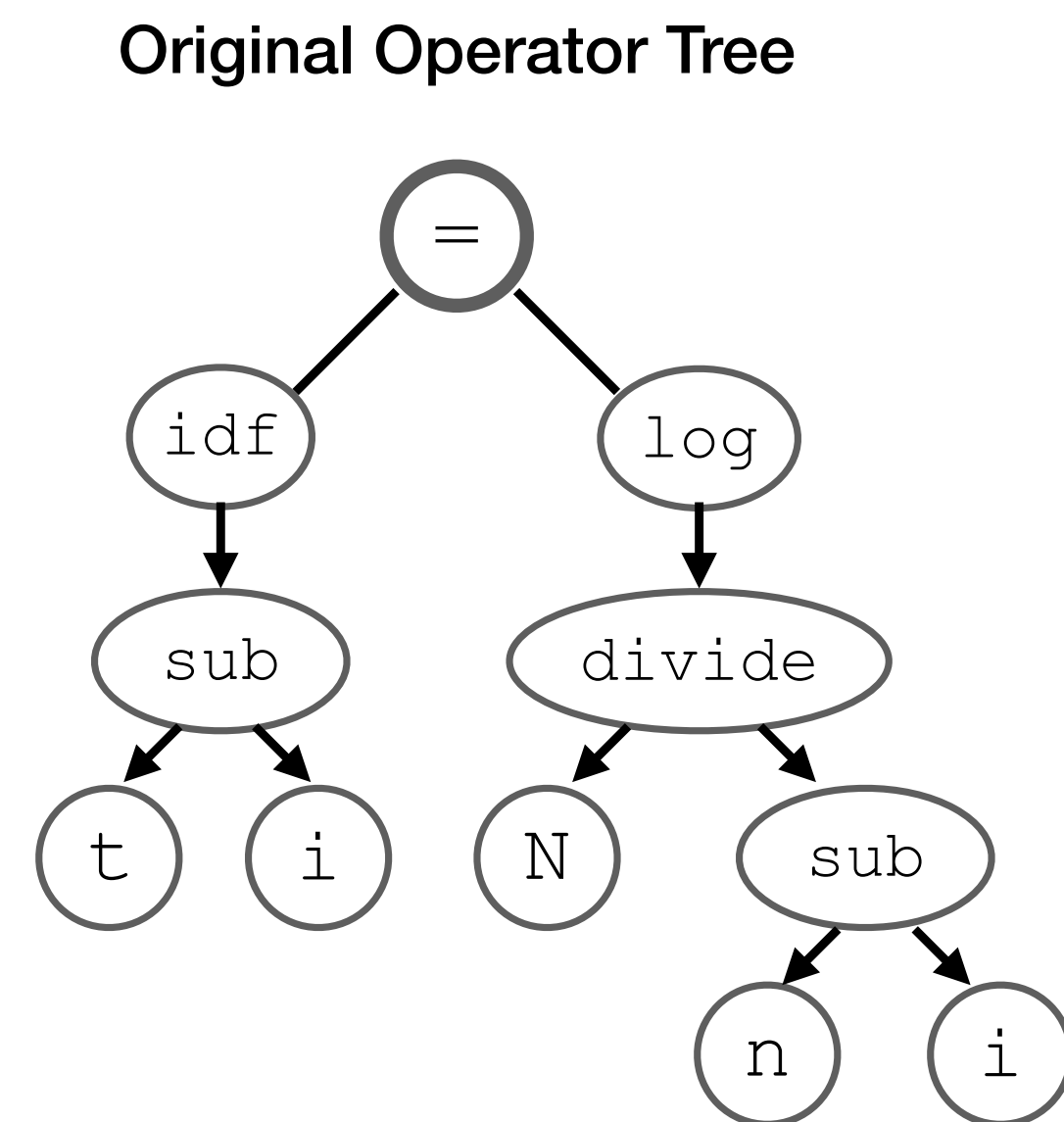


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Common Abstractions for Formula Search

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Example 2.10: OPT Variable Enumeration and Symbol Types

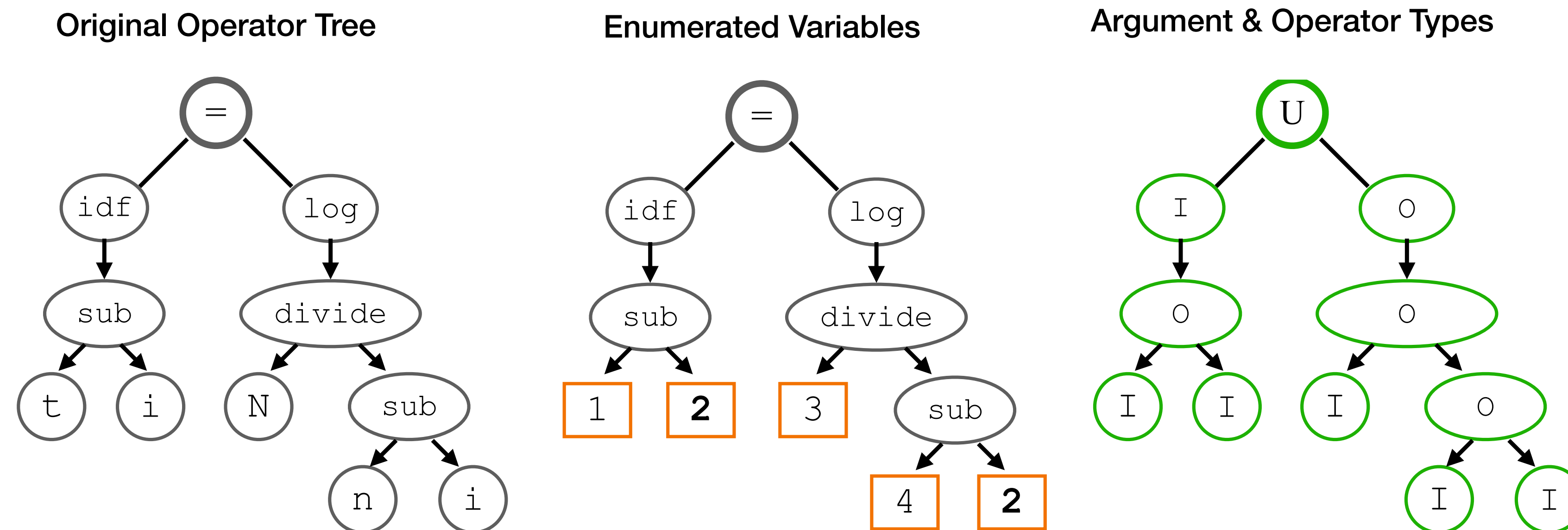


Assigning Types
Symbols/Nodes

Common Abstractions for Formula Search

Variable Enumeration and Types

Example 2.10: OPT Variable Enumeration and Symbol Types



U: Operator w.
unordered args

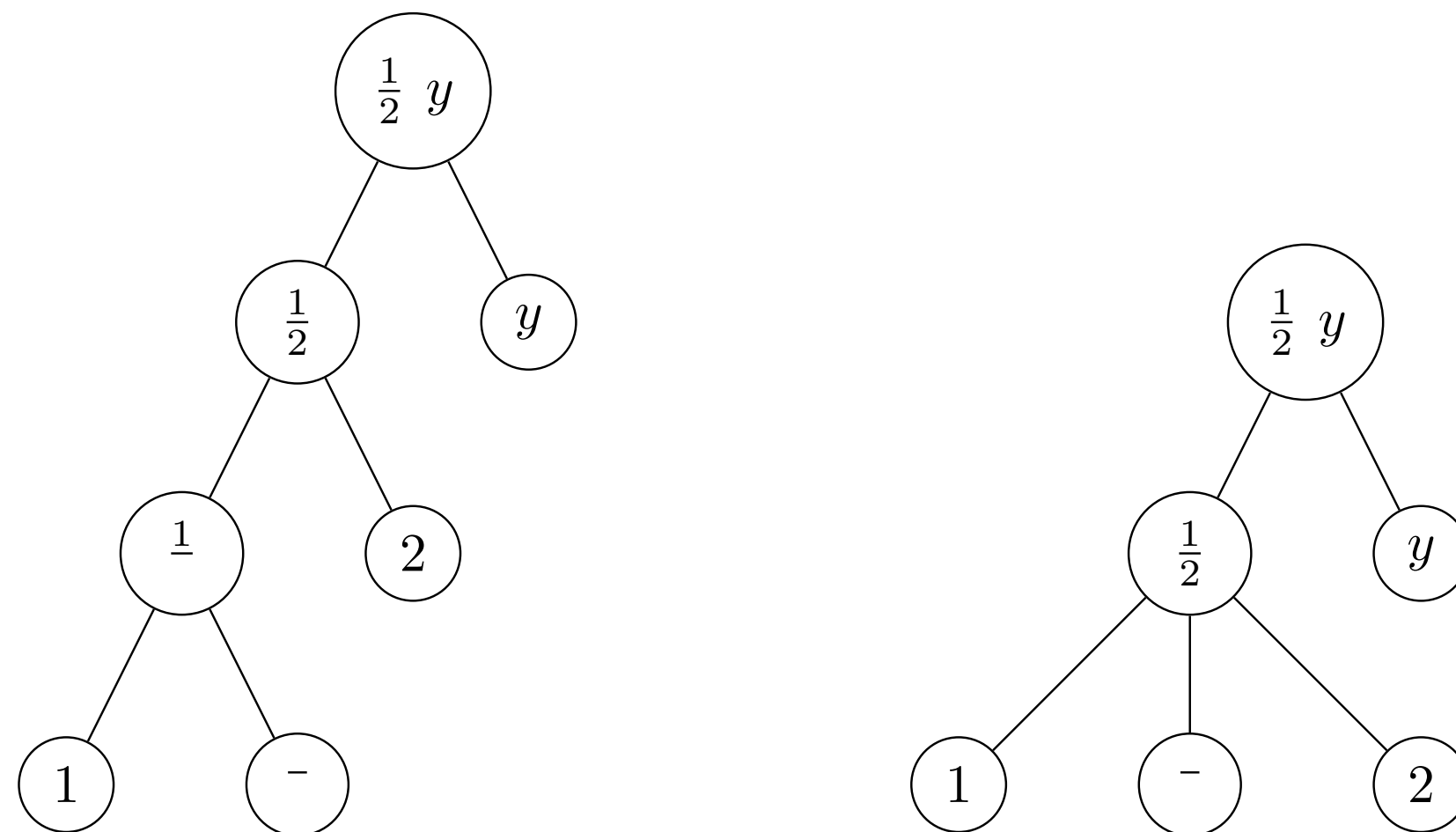
O: Operator w.
ordered args

I: Identifier

Spatial Formula Representations

Some examples

XY-Cut Trees (left: Recursive, right: Standard)



Pyramidal Histogram of Characters (XY-PHOC)

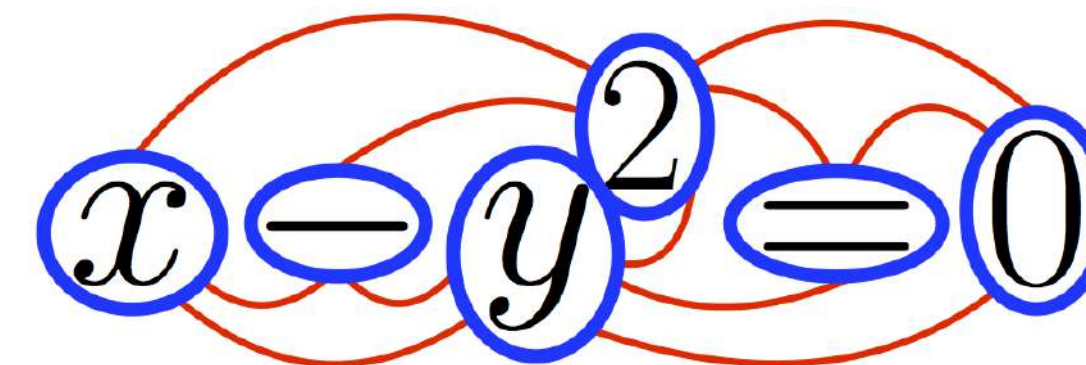
Level 1 $2y^8 = \sqrt{x}$

Level 2 $2y^8 \neq \sqrt{x}$

Level 2' ~~$2y^8 = \sqrt{x}$~~

Level 3 $2y^8 \mid = \mid \sqrt{x}$

Level 3' ~~$2y^8 = \sqrt{x}$~~



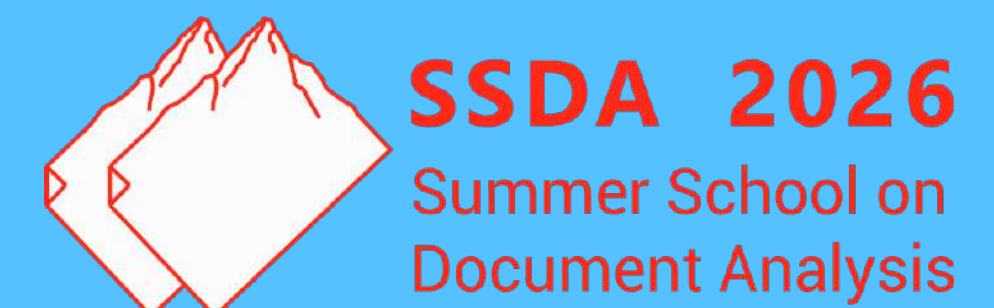
Appearance (LOS)

Tangent-V Demonstration:

Handwritten Formula Search & Content-Based Navigation in Lecture Videos

Kenny Davila , Richard Zanibbi :

Visual Search Engine for Handwritten and Typeset Math in Lecture Videos and LATEX Notes. ICFHR 2018: 50-55



View Lecture - NM_lecture_05

Normal

Binary

Content

Defn of Linear Span:
Let $S = \{\vec{v}_1, \dots, \vec{v}_n\}$ in $\mathbb{R}^n$

Linear span of S

$$L(S) = \{ \vec{w} \in \mathbb{R}^n \mid \vec{w} = \text{linear comb of vectors in } S \}$$

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$\mathbb{R}^2$
 $S = \{ \vec{v}_i \}$ $L(S) = \{$

Q2 Given a set of vectors describe geometric & algebraic of the span?

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4 / 21 - 00:09:59-00:11:33

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5 / 21 - 00:11:33-00:13:10

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6 / 21 - 00:13:10-00:14:24

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7 / 21 - 00:14:24-00:15:10

<< Prev

Next >>



View Lecture - NM_lecture_05

Normal

Binary

Content

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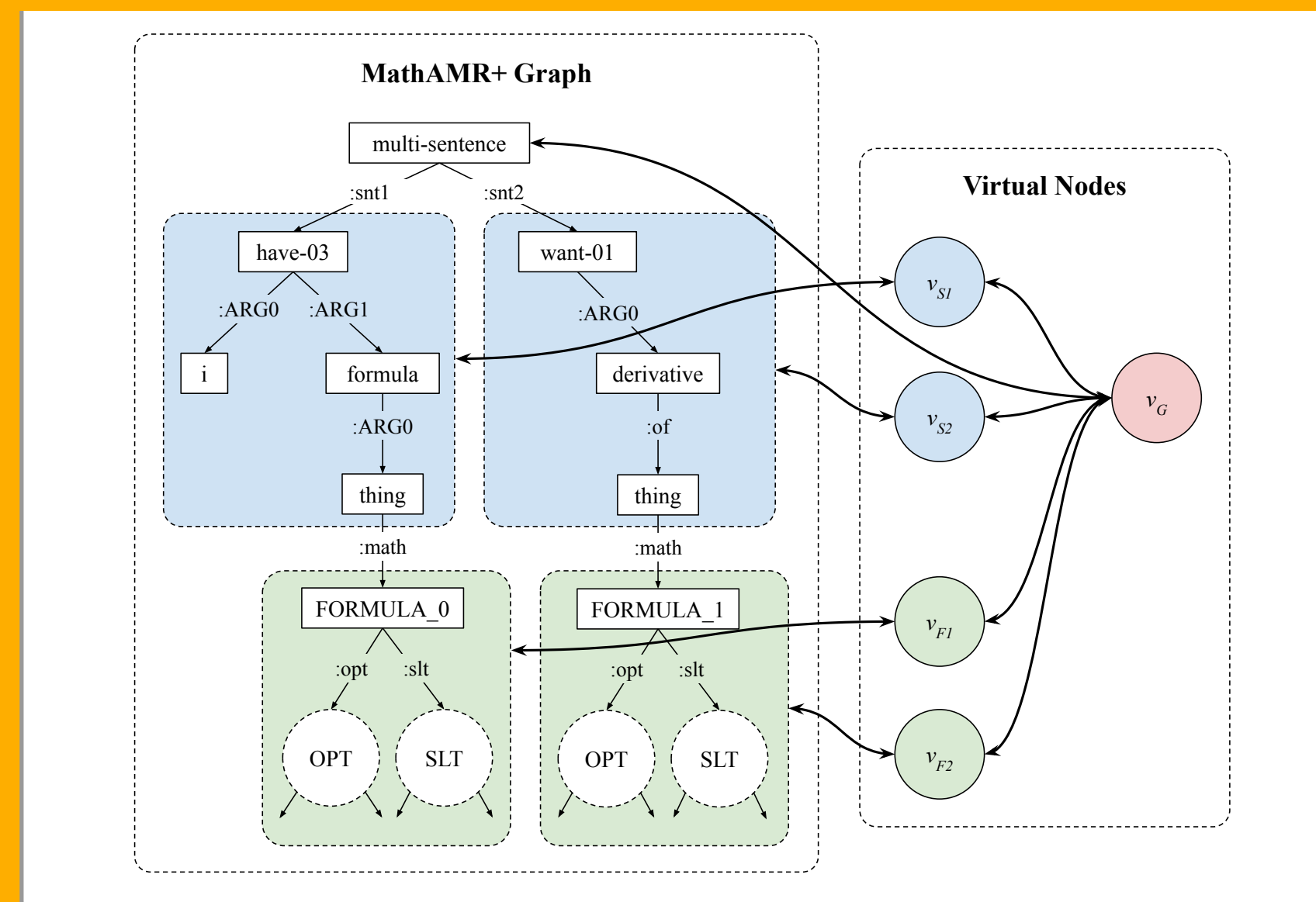
7 / 21 - 00:14:24-00:15:10

<< Prev

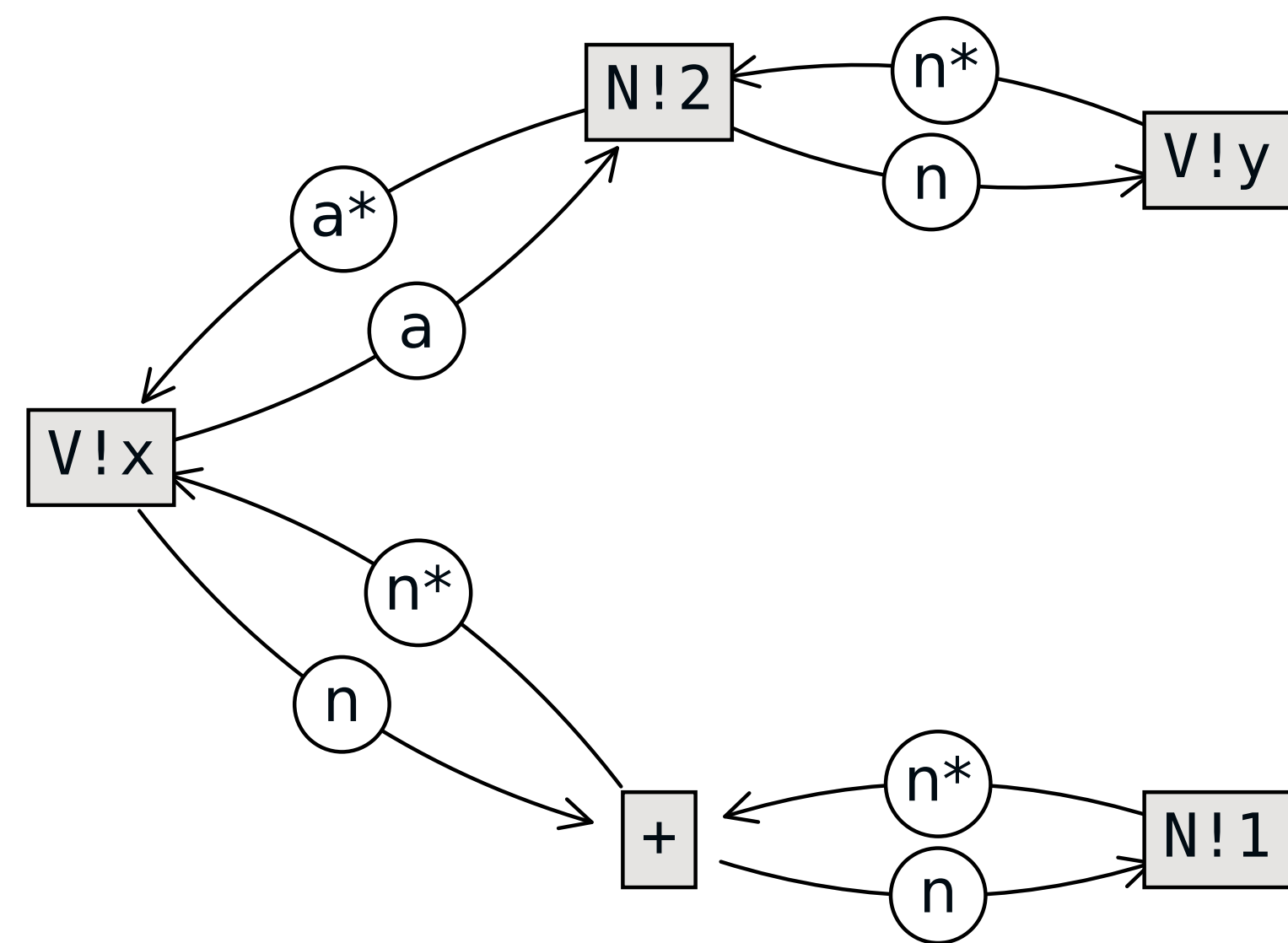
Next >>

Progress indicator: 19 dots, with the 10th dot highlighted in orange.

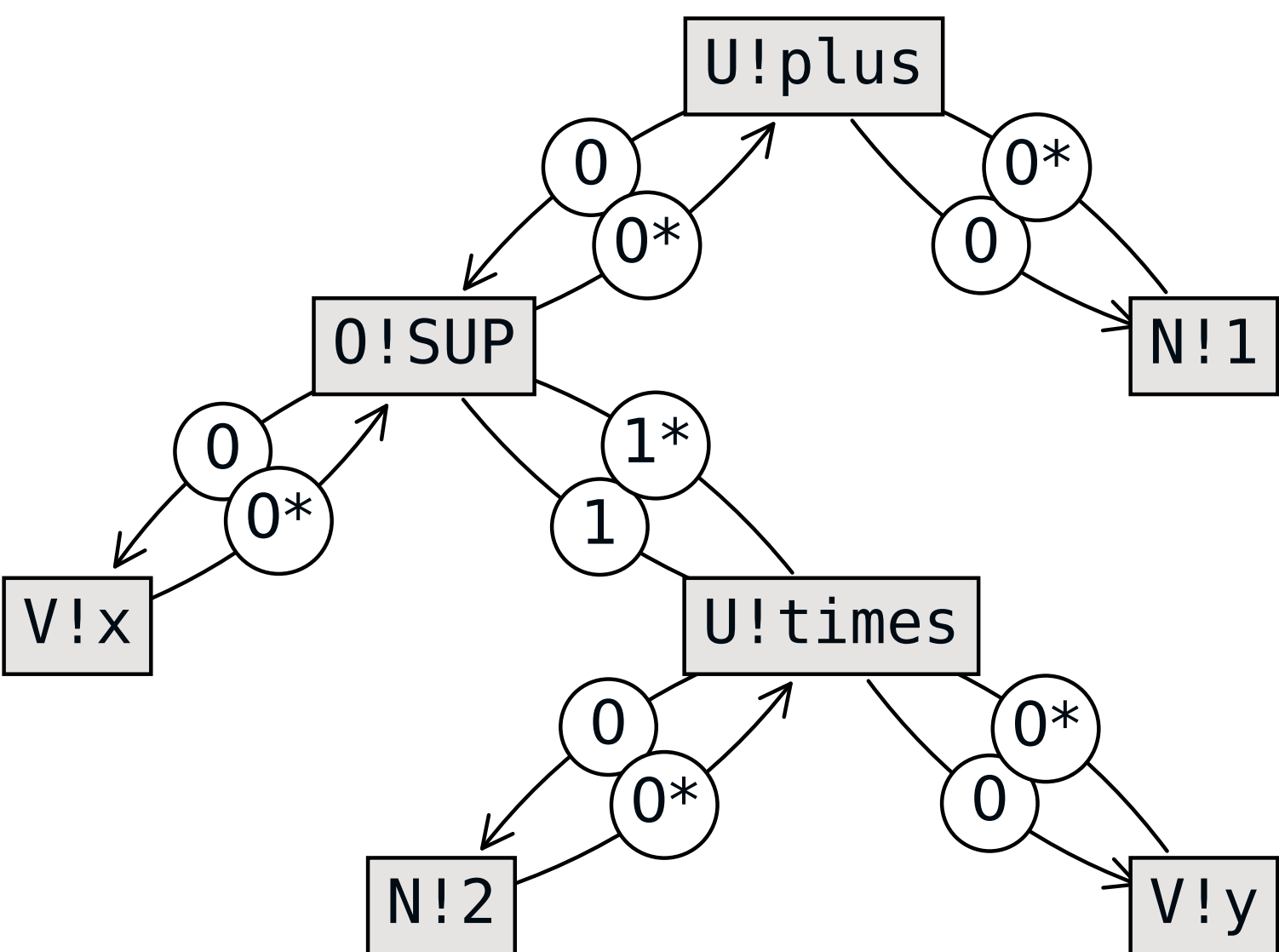
Graph Neural Networks for Formula & Text+Formula Retrieval



Node-Based Retrieval Using Graph Neural Nets

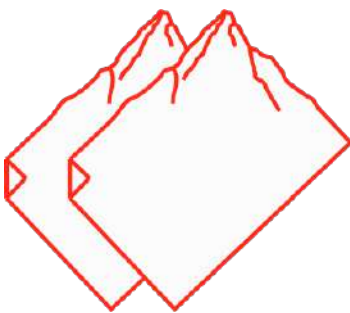


(a) Symbol Layout Graph (SLG)

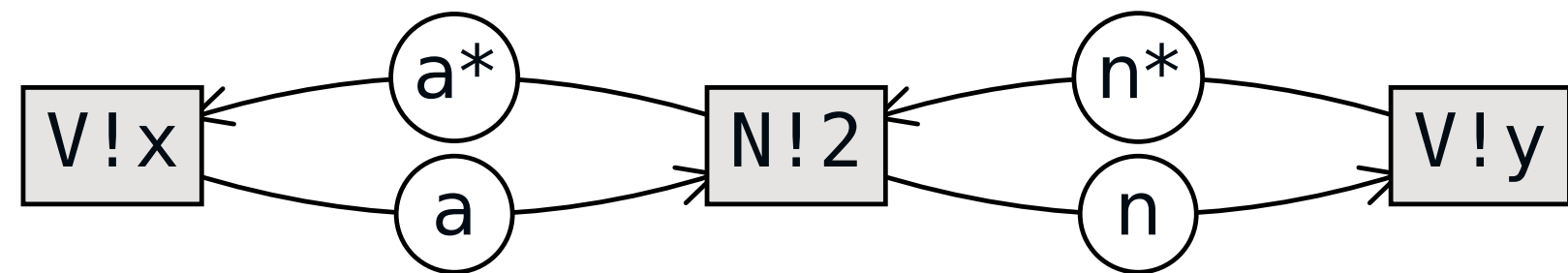


(b) Operator Graph (OPG)

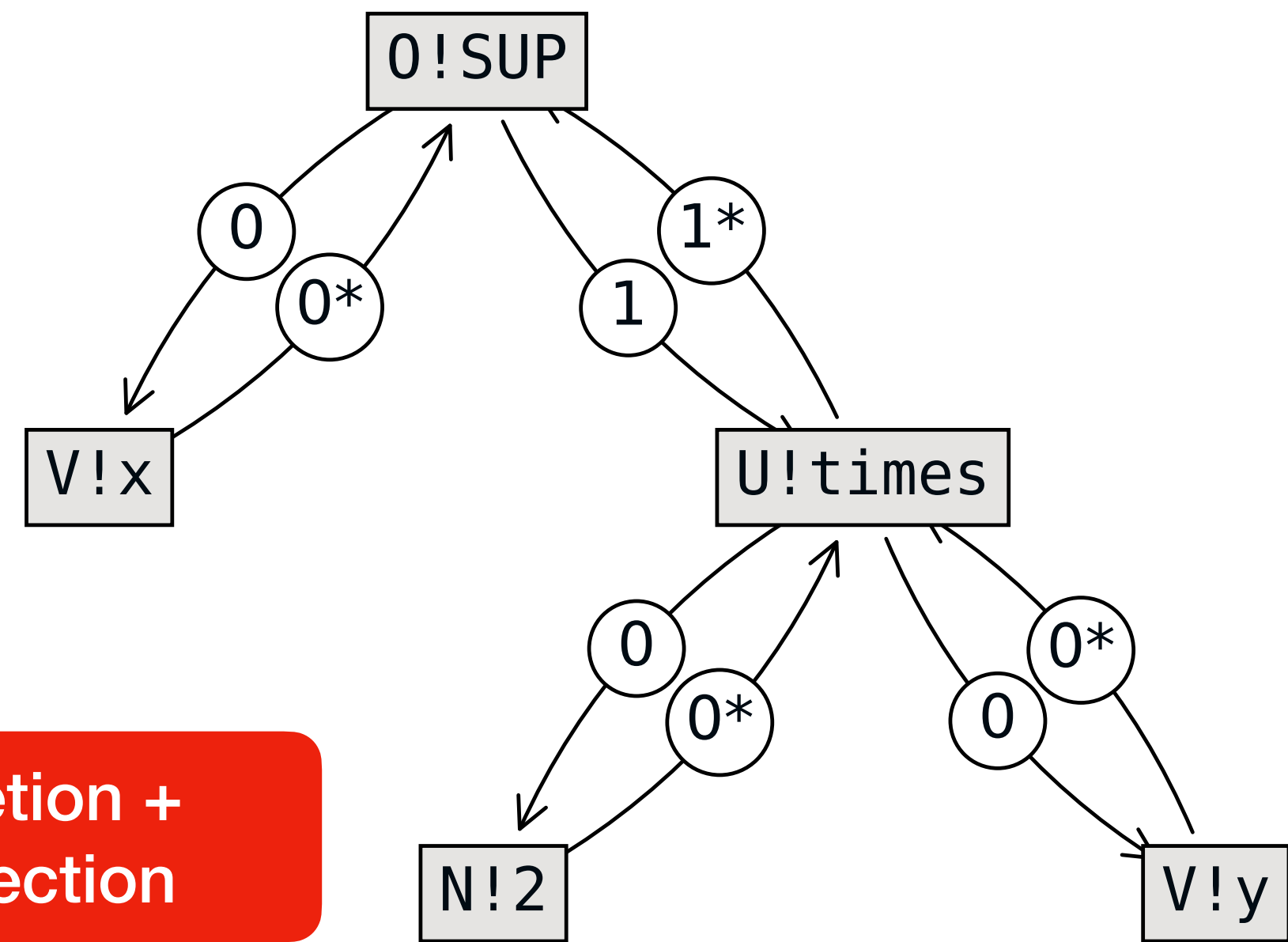
$$x^{2y} + 1$$



Node-Based Retrieval Using Graph Neural Nets



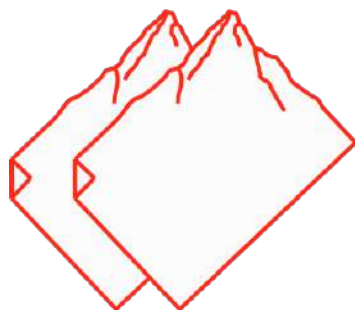
(c) SLG sub-expression



(d) OPG sub-expression

Random Node Deletion +
Larger Subtree Selection

$x^2y + 1$



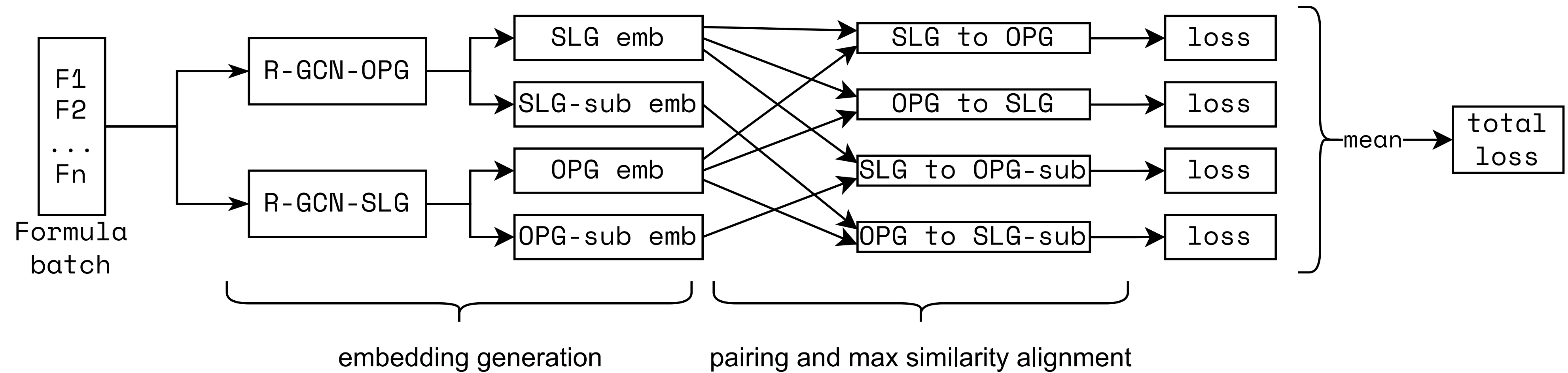
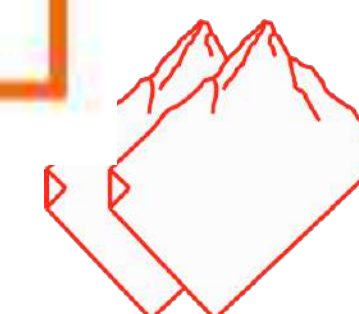


Figure 3: Training with contrastive learning across SLG and OPG nodes using the Outer set of graph augmentations. Positive and negative node pairs are created by maxsim alignment of nodes from the first to second type in each graph pairing. This occurs both for complete formulas (i.e., whole OPGs/SLGs), and for complete formulas in one representation vs. sub-expressions in the other representation (e.g., (SLG, OPG-sub)). The final loss is averaged across the four graph type pairings.

Table 2. ARQMath-3 Formula Search Task Results (76 Test Queries).

Approach0 is a state-of-the-art formula retrieval model. Text + formula search results are shown for context

Model	NDCG'	MAP'	P'@10
Formula Search			
Approach-0 [4]	0.639	0.501	0.615
Inner+Outer MAX (ours)	0.701	0.505	0.597
<i>*Inner RRF (ours, <u>updated</u>)</i>	<u>0.773</u>	<u>0.554</u>	<u>0.632</u>
Text + Formula Search			
Approach0+ColBERT[4]	0.720	0.568	0.688
TanCFT MathAMR [5]	0.640	0.388	0.478



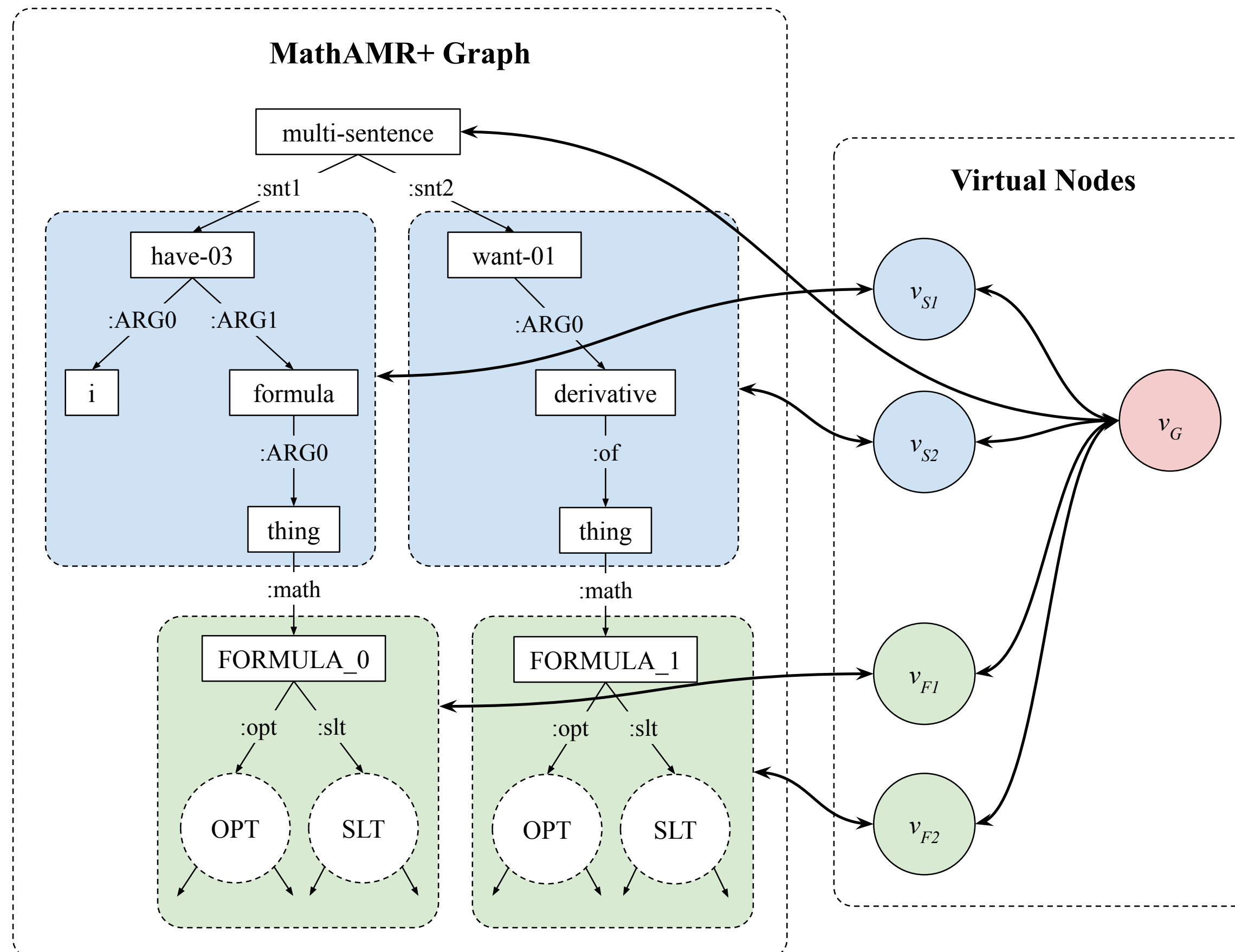
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Moving to Text + Formulas

Original Question

“I have the formula $f(x) = x^2 + 1$. What is the derivative of $f(x)$?”

MathAMR+ Graph with Hierarchical Virtual Nodes



Using **Abstract Meaning Representation (AMR)** graphs produced by encoder-decoder

Formulas replaced by identifiers before AMR generation;
OPT + SLT added as id children

Addition of **virtual nodes** for:

- Whole post
- Each sentence
- Each formula

Results on (Multimodal) Answer Retrieval

ARQMath-3 Benchmark

Table 4.19: ARQMath-3 Answer Retrieval (Task 1) benchmarking results.

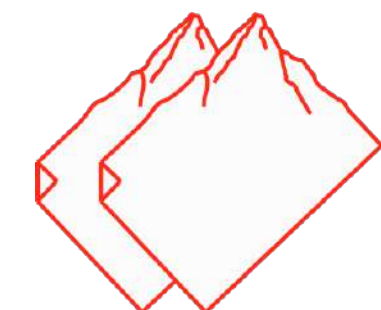
System	nDCG'	MAP'	P'@10	Bpref
Approach0 / Struct.+ColBERT (Coco-MAE) [68]	0.546	0.237	0.360	0.221
Kassaie et al. / RRF+LLM [23]	0.522	0.195	0.277	—
MSM / Ensemble RRF [14]	0.504	0.157	0.241	0.138
MIRMU / MiniLM+RoBERTa [50]	0.498	0.184	0.267	0.169
Satpute et al. / GPT-4+DPR [52]	0.486	0.219	0.374	0.225
DPRL / QQ-QA-AMR [36]	0.185	0.040	0.091	—
Ours / MathAMR+	0.334	0.108	0.236	0.163

Embed size: 64
Batch size: 16

Jacob Yoon.

MathAMR+: A Unified Graph Neural Network Framework for Multimodal Mathematical Information Retrieval.

MSc Thesis, RIT, 2026.



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Results for Contextualized Formula Retrieval

ARQMath-3 Benchmark

Table 4.21: ARQMath-3 Formula Retrieval (Task 2) benchmarking results.

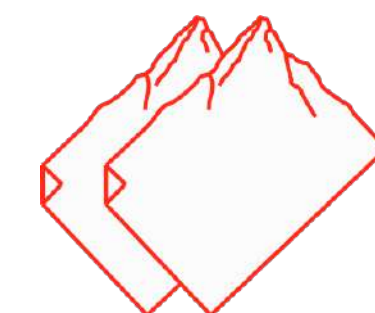
System	nDCG'	MAP'	P'@10	Bpref
L2L / All (Unweighted) [60]	0.849	0.620	0.680	—
Amador et al. / Inner RRF [†] [4]	0.773	0.554	0.632	—
Approach0 / Struct.+ColBERT [68]	0.720	0.568	0.688	0.560
Amador et al. / In+Out [4]	0.701	0.505	0.597	—
DPRL / TanCFT+MathAMR [36]	0.681	0.471	0.617	—
MathDowers / latex_L8_a040 [20]	0.640	0.451	0.549	0.443
DPRL / MathAMR ^{†1} [36]	0.579	0.367	0.549	—
DPRL / MathAMR ² [35]	0.316	0.160	0.253	—
Embed size: 64 Batch size: 16 Ours / MathAMR+	0.507	0.333	0.588	0.503

[†] Unofficial run by the authors

Jacob Yoon.

MathAMR+: A Unified Graph Neural Network Framework for Multimodal Mathematical Information Retrieval.

MSc Thesis, RIT, 2026.

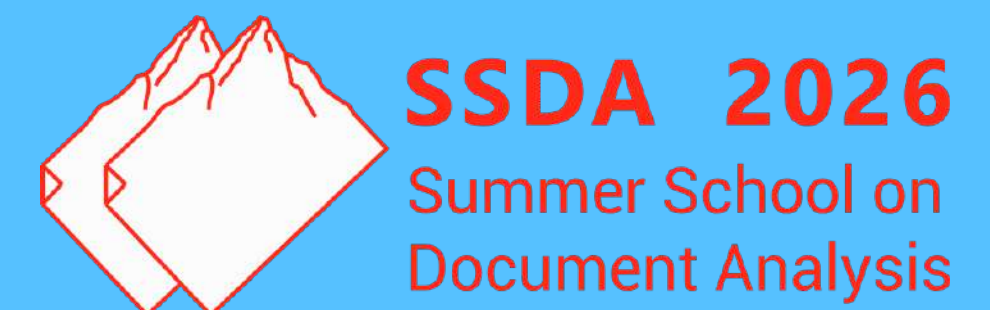


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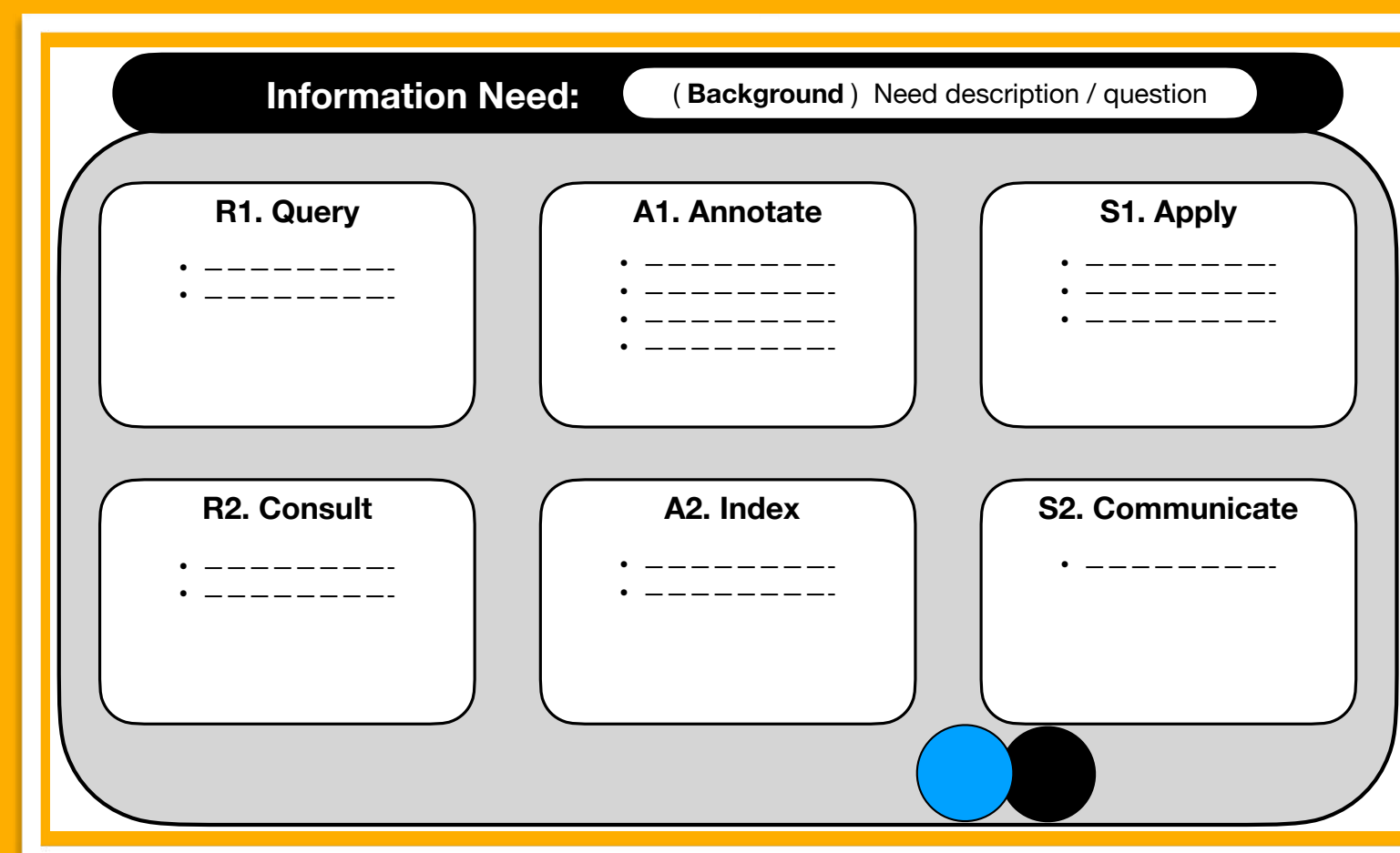
MathDeck Demonstration:

Multimodal Search in PDF Documents

Bryan Amador , Matt Langsenkamp , Abhisek Dey , Ayush Kumar Shah , Richard Zanibbi 
Searching the ACL Anthology with Math Formulas and Text. SIGIR 2023: 3110-3114



The 'Task Jar' Framework for Human Retrieval: Needs, Tasks, and Sources



Information Needs

An Example from Math

Example 1.1: Differing Information Needs

Query: What does $a^2 + b^2 = c^2$ represent and how is it useful?

Students

Educators

Researchers

Information need is closely related to the **query intent**, i.e., which information a user hopes a prompt will retrieve

The background of the searcher strongly impacts relevance, in terms of **both pertinence, and interpretability**

Information Needs

An Example from Math

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Students might use this query to learn the Pythagoras theorem, and perhaps find an example demonstrating the theorem, and a possible proof.

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Information Needs

An Example from Math

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Educators may have similar interests to students, but may seek additional resources on how to teach this result.

Researchers

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Information Needs

An Example from Math

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Students might use this query to learn the Pythagoras theorem, and perhaps find an example demonstrating the theorem, and a possible proof.

Educators may have similar interests to students, but may seek additional resources on how to teach this result.

Researchers can have very different interests than the other audiences. They may be interested in one or more of the following:

- For a mathematician: Is this true in a general metric space and/or a Hilbert space?
- For a physicist: How is it linked to the probability assignments in quantum mechanics?
- For an IR expert: How is it related to probability assignments in a Hilbert space used in describing interaction for information retrieval?

Information need is closely related to the **query intent**, i.e., which information a user hopes a prompt will retrieve

The background of the searcher strongly impacts relevance, in terms of **both pertinence, and interpretability**

Information Task Framework

Overview

Retrieve

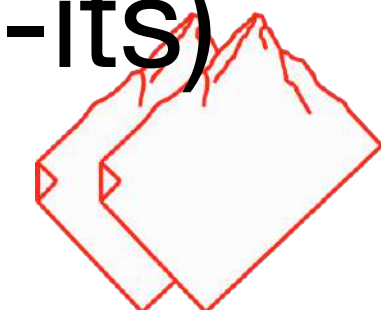
Analyze

Synthesize

Retrieve - Analyze - Synthesize

Adapted from **sense-making**, i.e., organizing information on a large topic (e.g., for a term paper on IDF and its applications)

Often, when we find or create information we **create information sources of our own** (e.g., papers, post-its)



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Information Task Framework

Overview

Retrieve Information

R1 **Query** to request sources of information

R2 **Consult** and interpret available sources, *examining* and *navigating* within and across sources

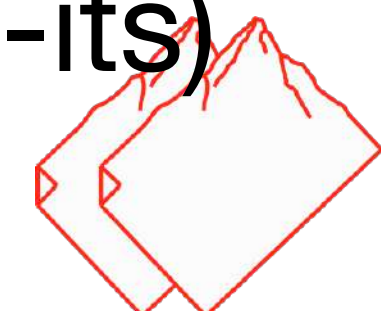
Analyze

Synthesize

Retrieve - Analyze - Synthesize

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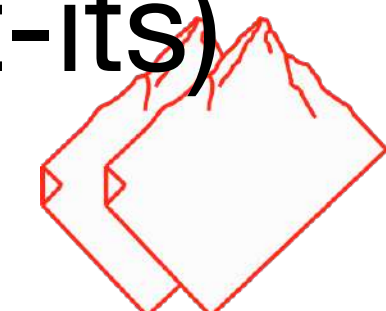
Analyze Information

- A1 **Annotate** sources with additional information, *e.g.*, notes, add formula locations
- A2 **Index** sources by organizing them for retrieval

Synthesize

Retrieve - Analyze - Synthesize
Adapted from **sense-making**, i.e., organizing information on a large topic (e.g., for a term paper on IDF and its applications)

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Information Task Framework

Overview

Retrieve Information

- R1 **Query** to request sources of information
- R2 **Consult** and interpret available sources, *examining* and *navigating* within and across sources

Analyze Information

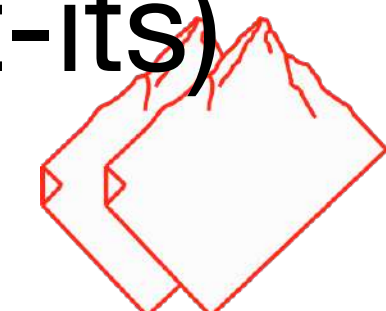
- A1 **Annotate** sources with additional information, *e.g.*, notes, add formula locations
- A2 **Index** sources by organizing them for retrieval

Synthesize Information

- S1 **Apply** available information that we know, have in available sources, or is encoded in algorithms, etc.
 - S2 **Communicate** information by creating new sources
-

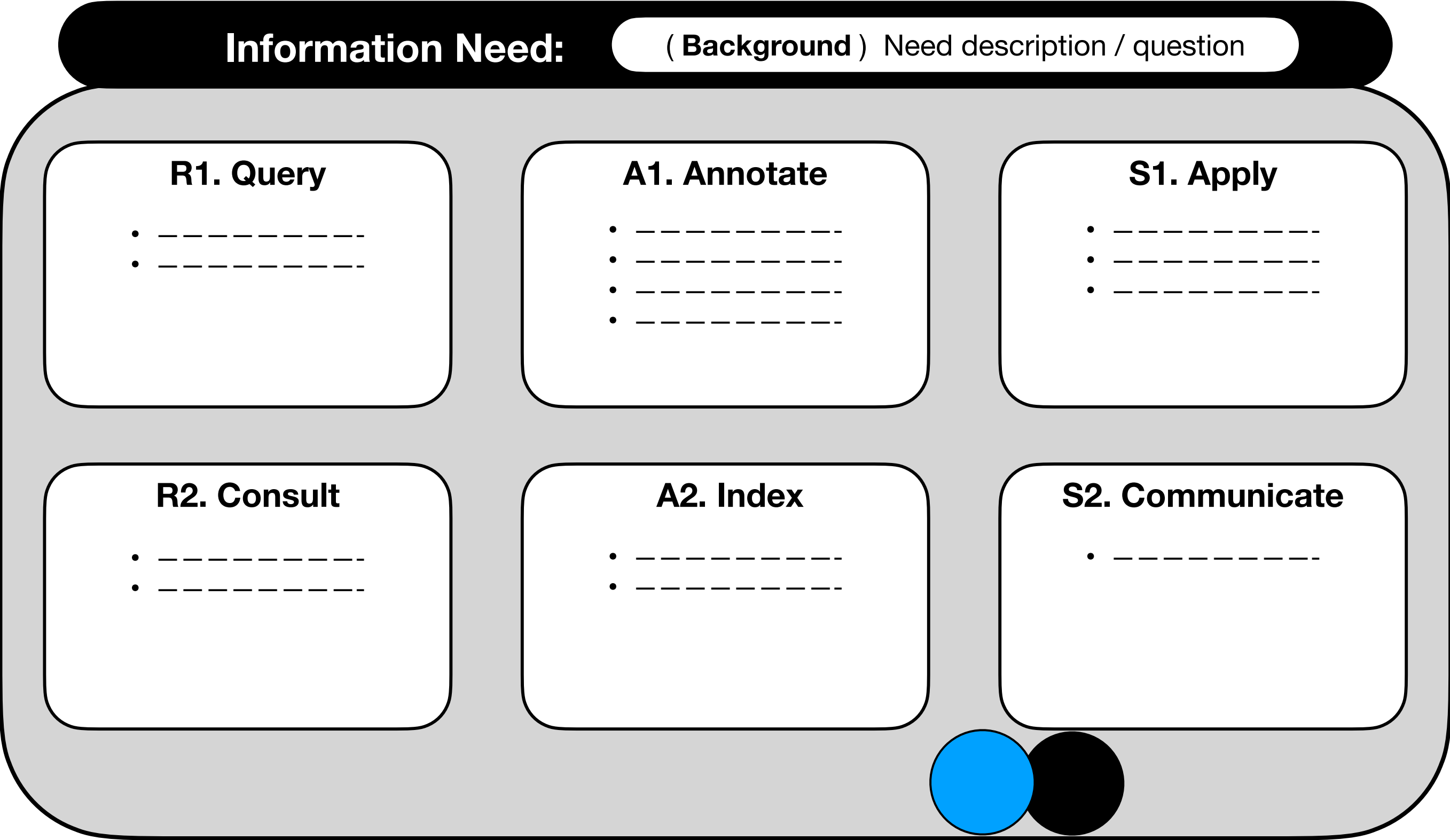
Retrieve - Analyze - Synthesize
Adapted from **sense-making**, i.e., organizing information on a large topic (e.g., for a term paper on IDF and its applications)

Often, when we find or create information we **create information sources of our own** (e.g., papers, post-its)

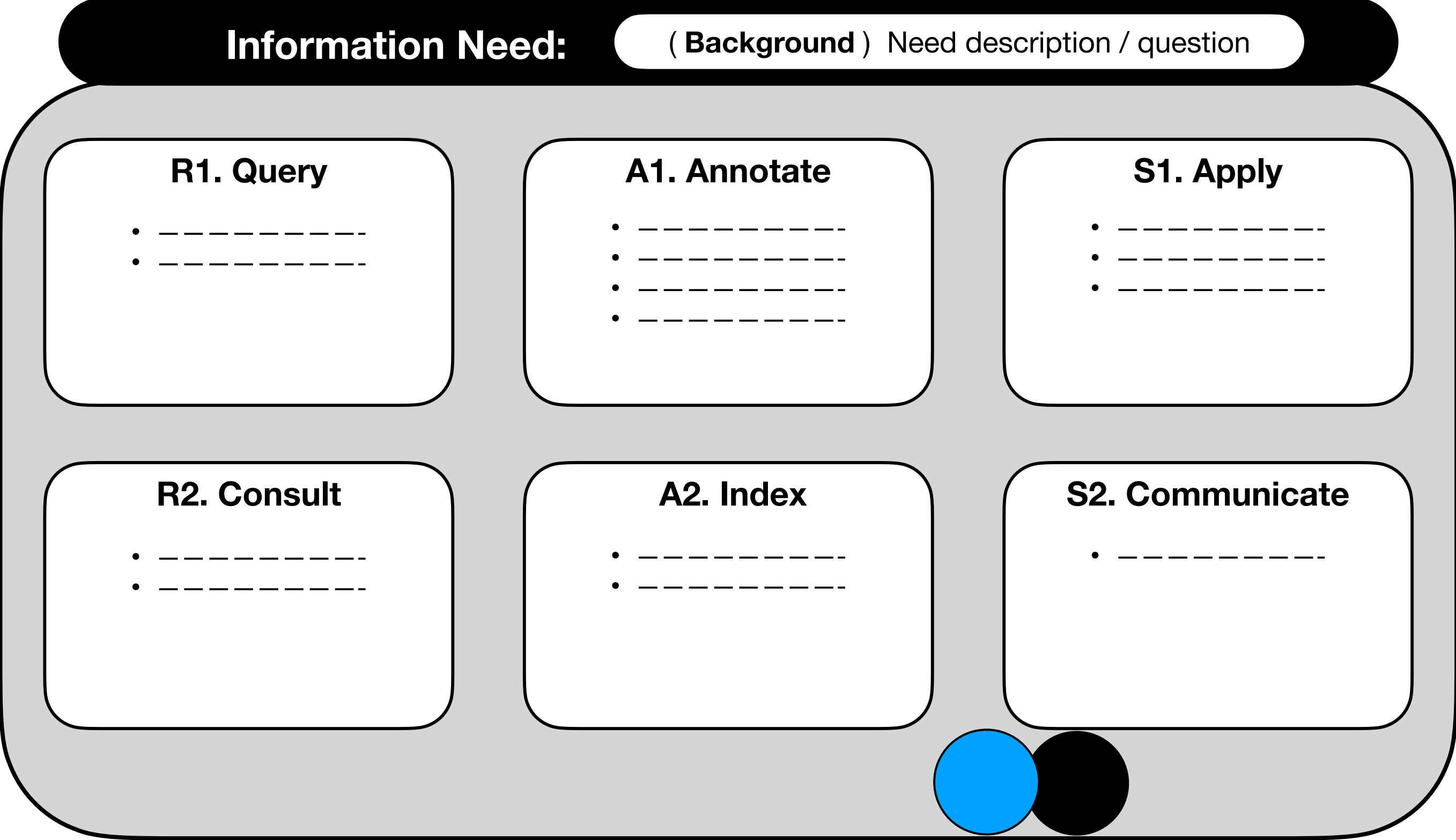


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Information Need & Task Interaction: The “Task Jar”

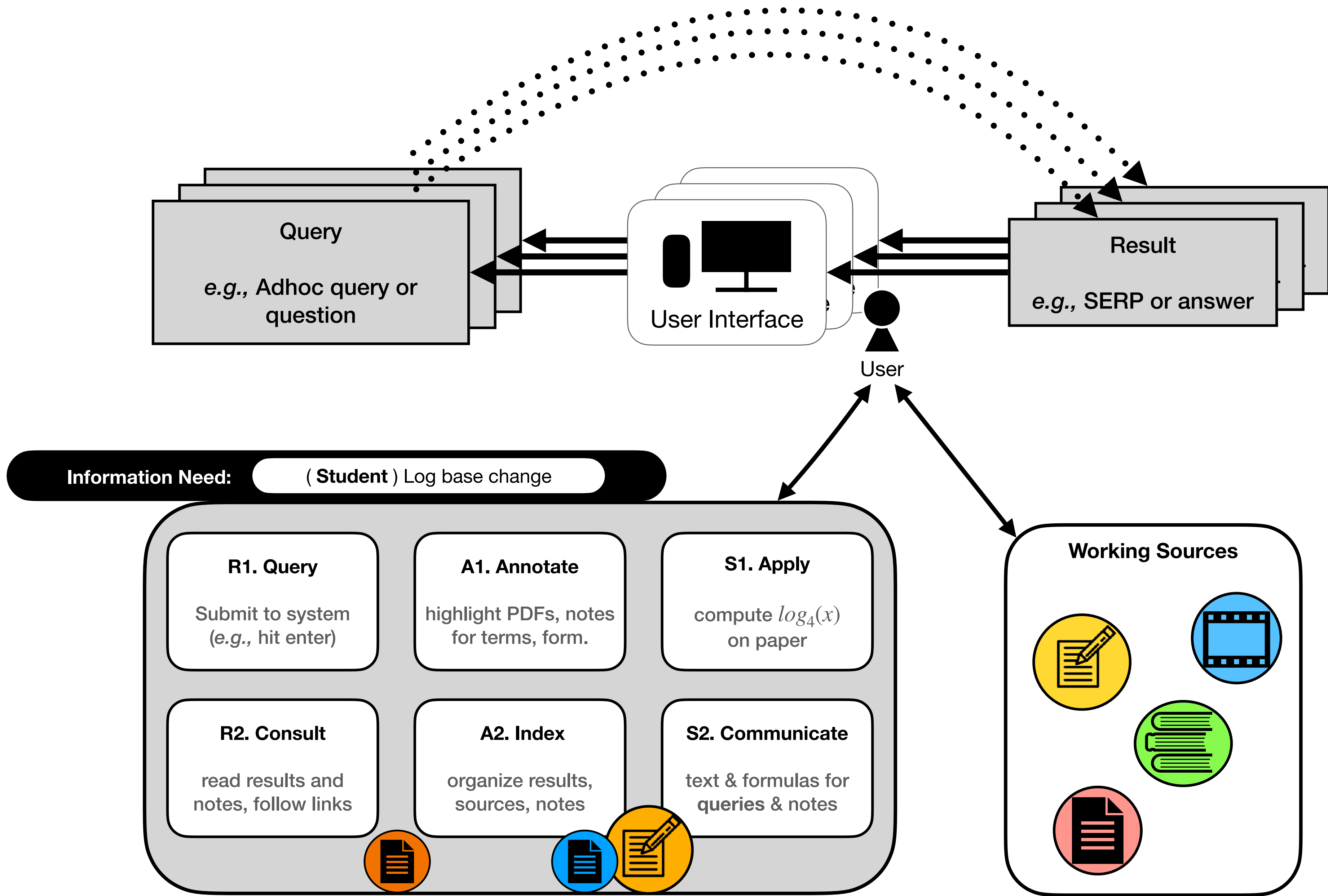


Information Need & Task Interaction: The “Task Jar”

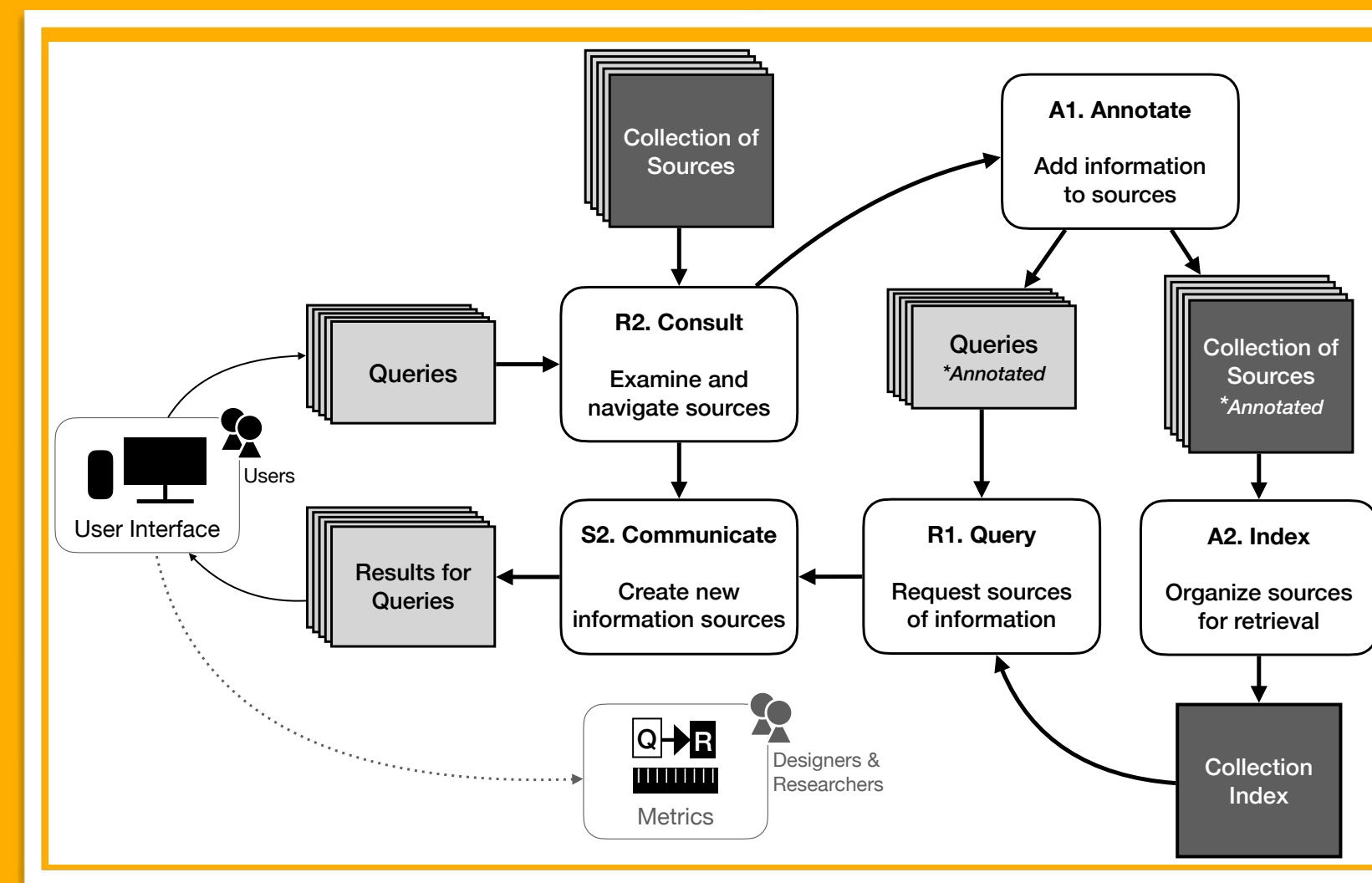


Information sources used in the wild by people are of many different types, e.g.,

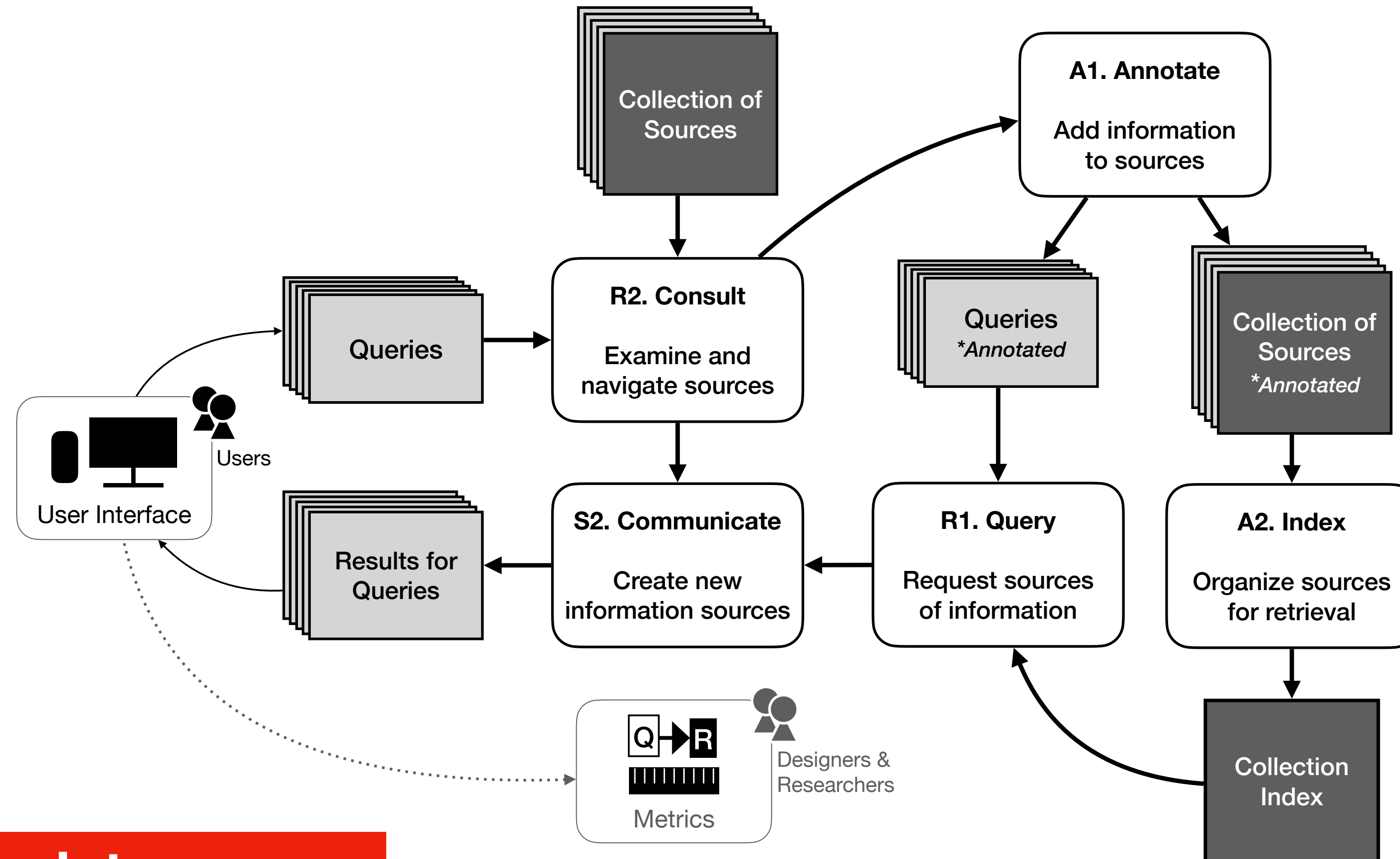
- Books, papers, Video, Audio
- **Queries**
- **Search results & LLM responses**
- Post-its (‘sticky note’)
- Conversations
- Observations
- **Formula chips & cards**
- **Generated whiteboard keyframes**



Related Information Task Model for Systems

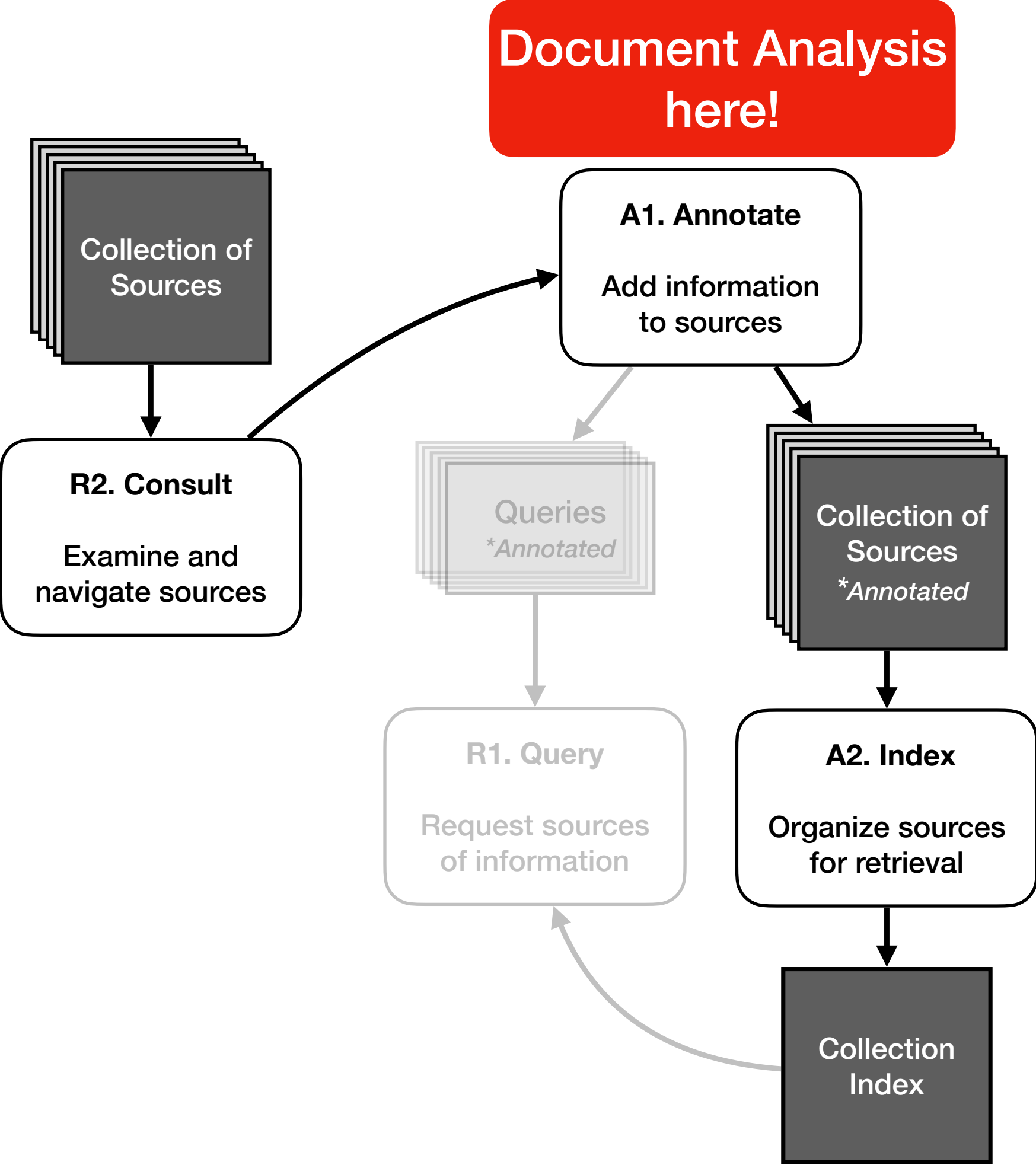


System Retrieval Model



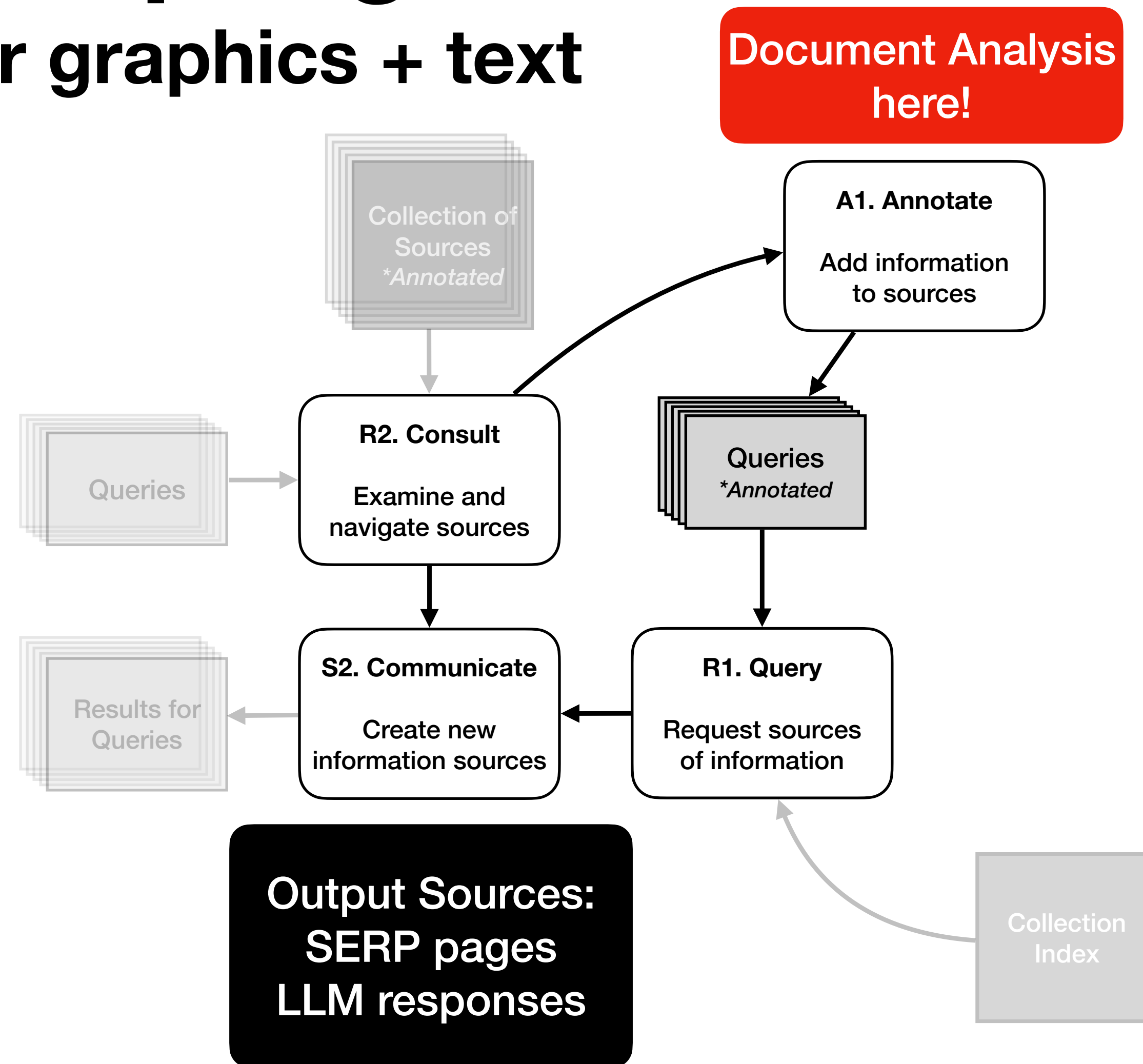
Note that the task types are same as for human searchers

Annotation and Indexing

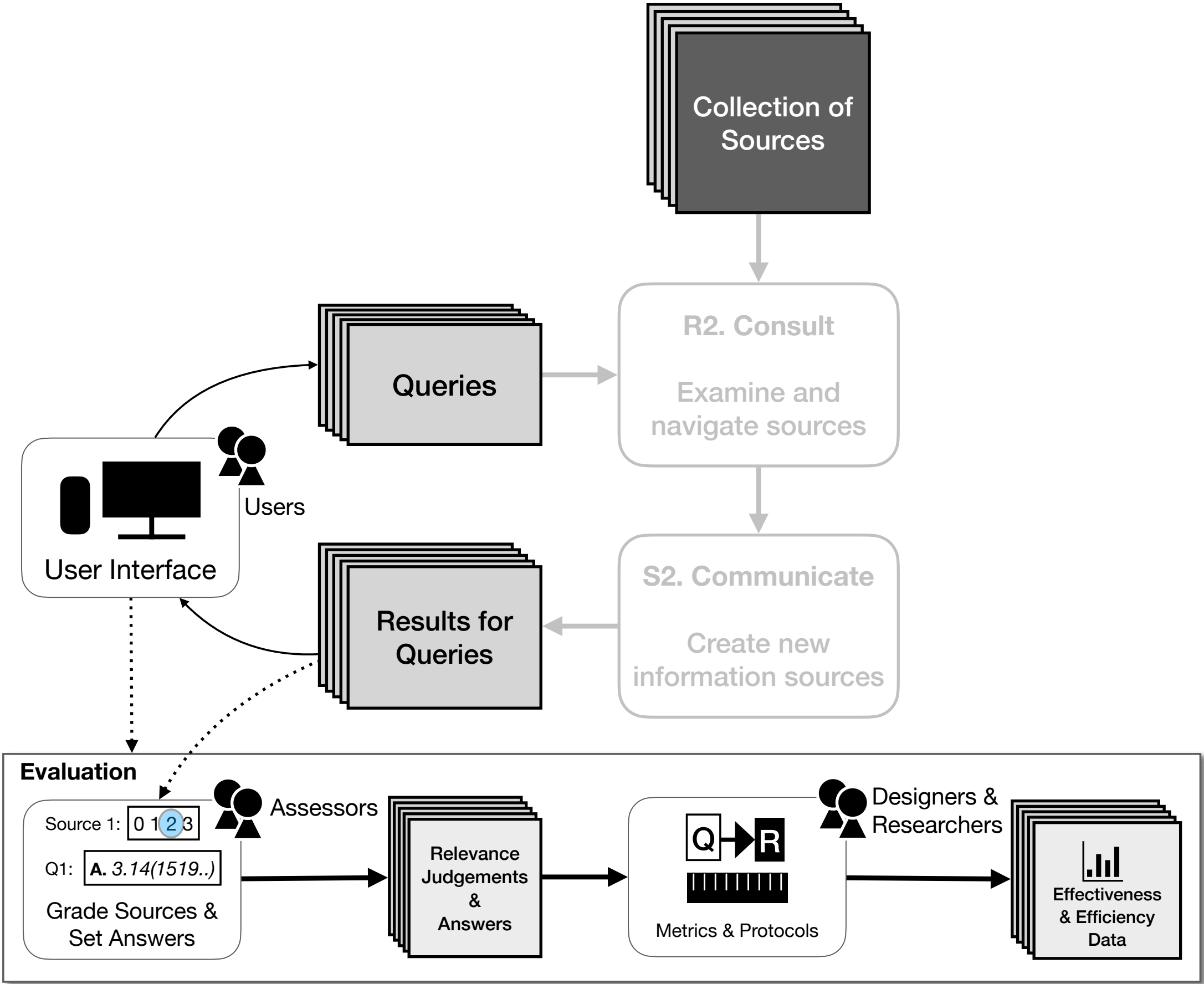


Retrieval / Prompting

e.g., formula, text, or graphics + text

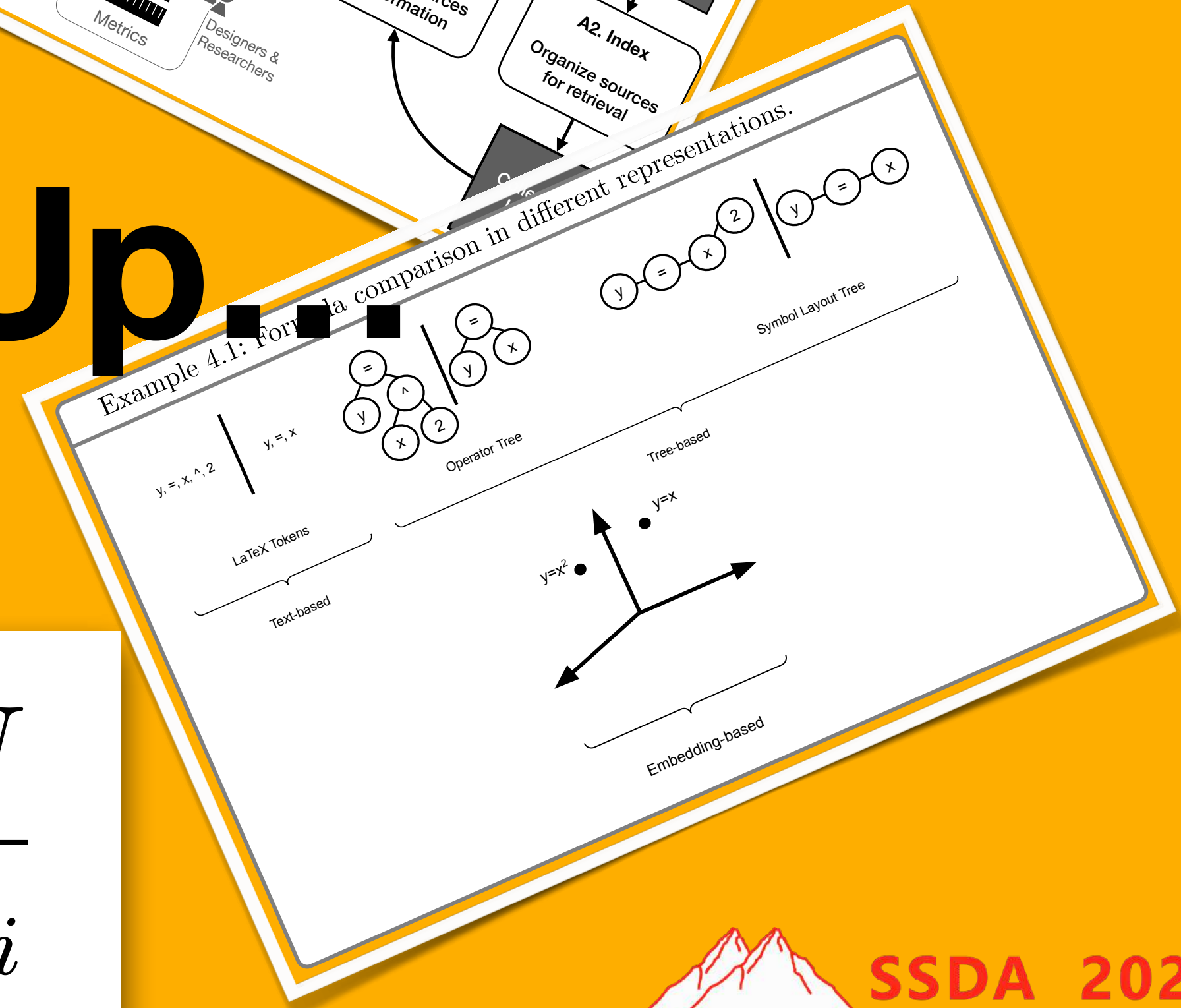
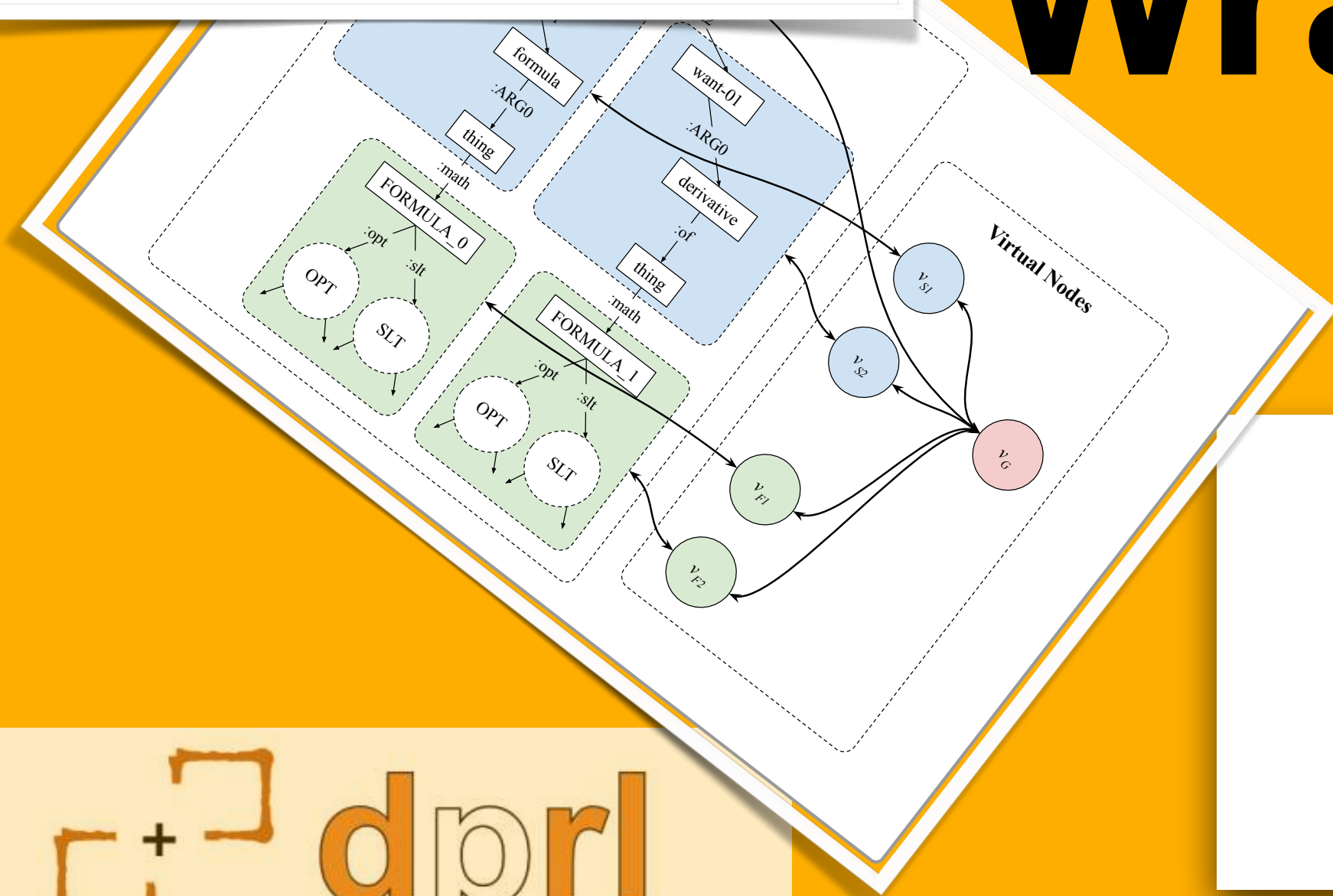
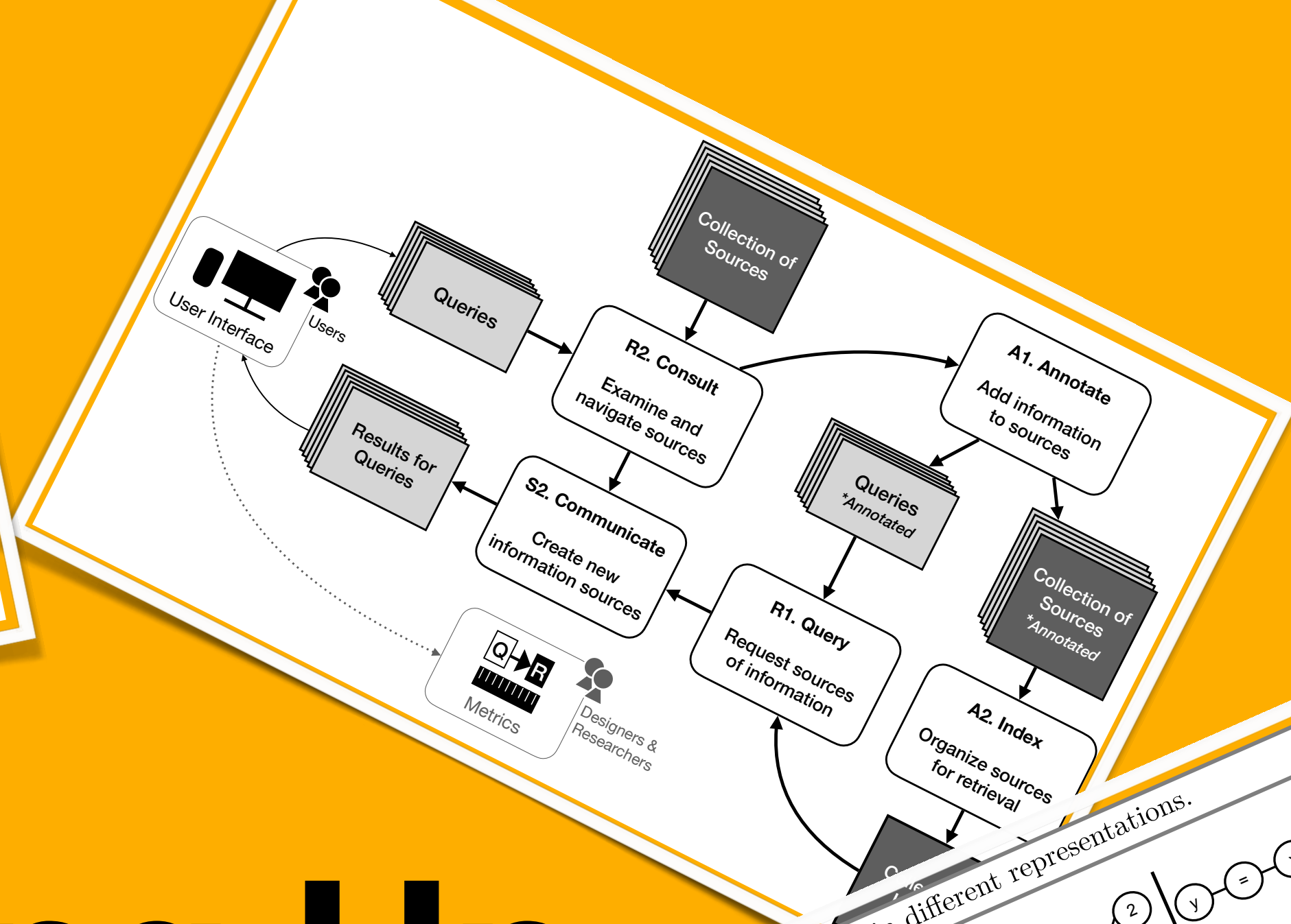
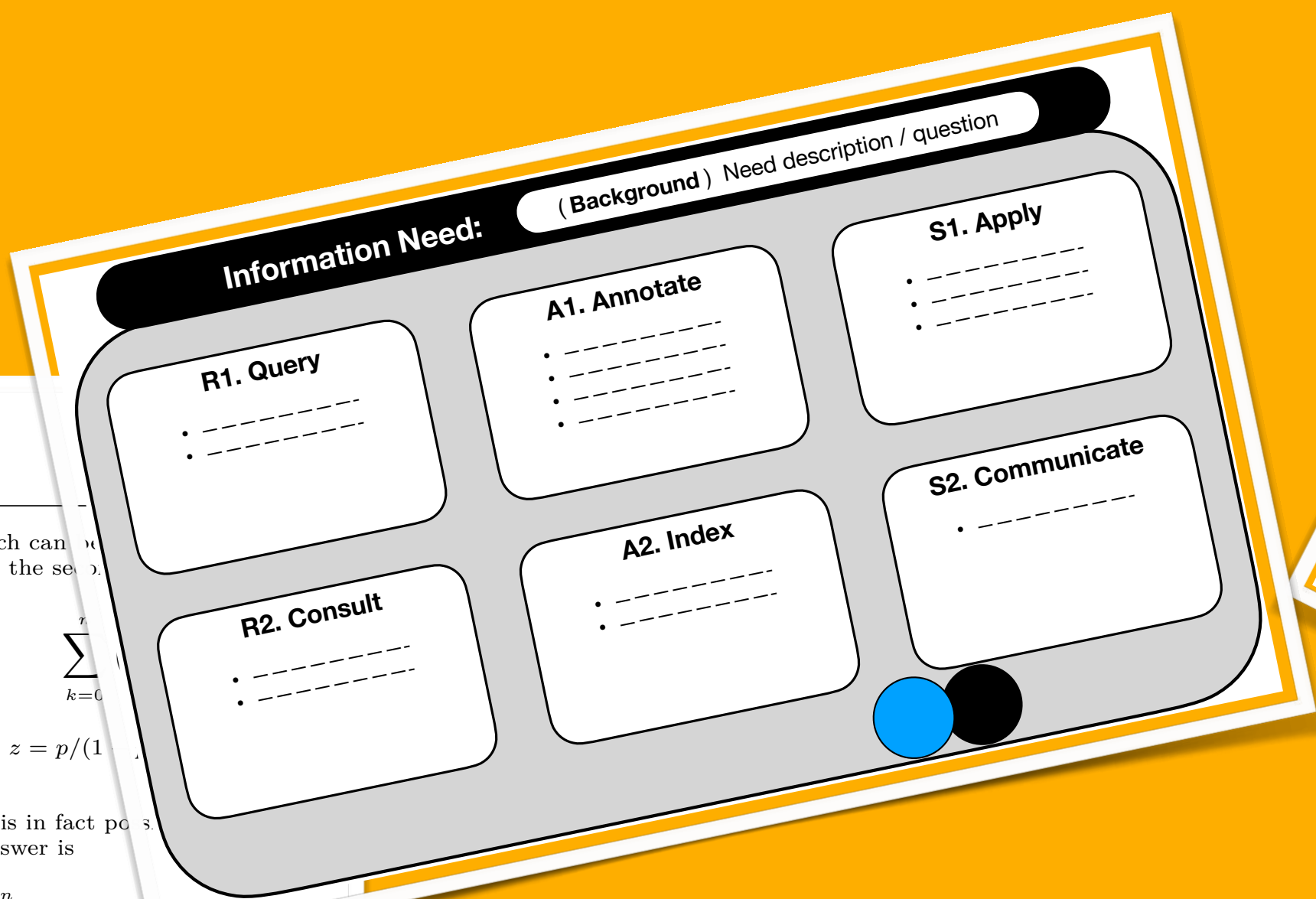


Evaluation



Math-aware search (ad-hoc retrieval)

Query	Result
I have the sum $\sum_{k=0}^n \binom{n}{k}$ know the result is $n^2 - 1$ but I don't know how you get there. How does one even begin to simplify a sum like this that has binomial coefficients.	1 ... which can be used in calculating the sequence $\sum_{k=0}^n \binom{n}{k}$ and let $z = p/(1 - p)$ 2 Yes, it is in fact possible to find the answer is $\sum_{k=0}^n \binom{n}{k} \binom{m}{k} = \binom{n+m}{n}$ assuming that $n \leq m$. This comes from the fact that ...



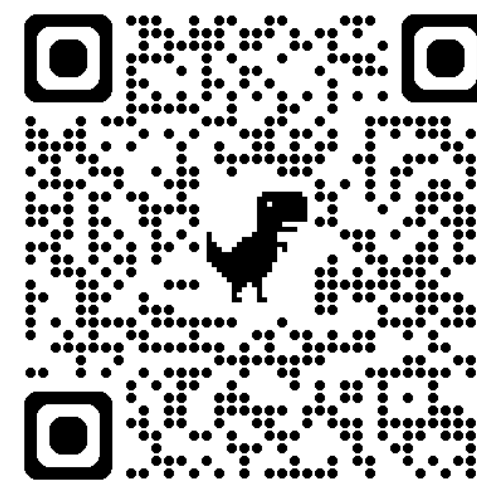
$$idf(t_i) = \log \frac{N}{n_i}$$

What We've Discussed Today

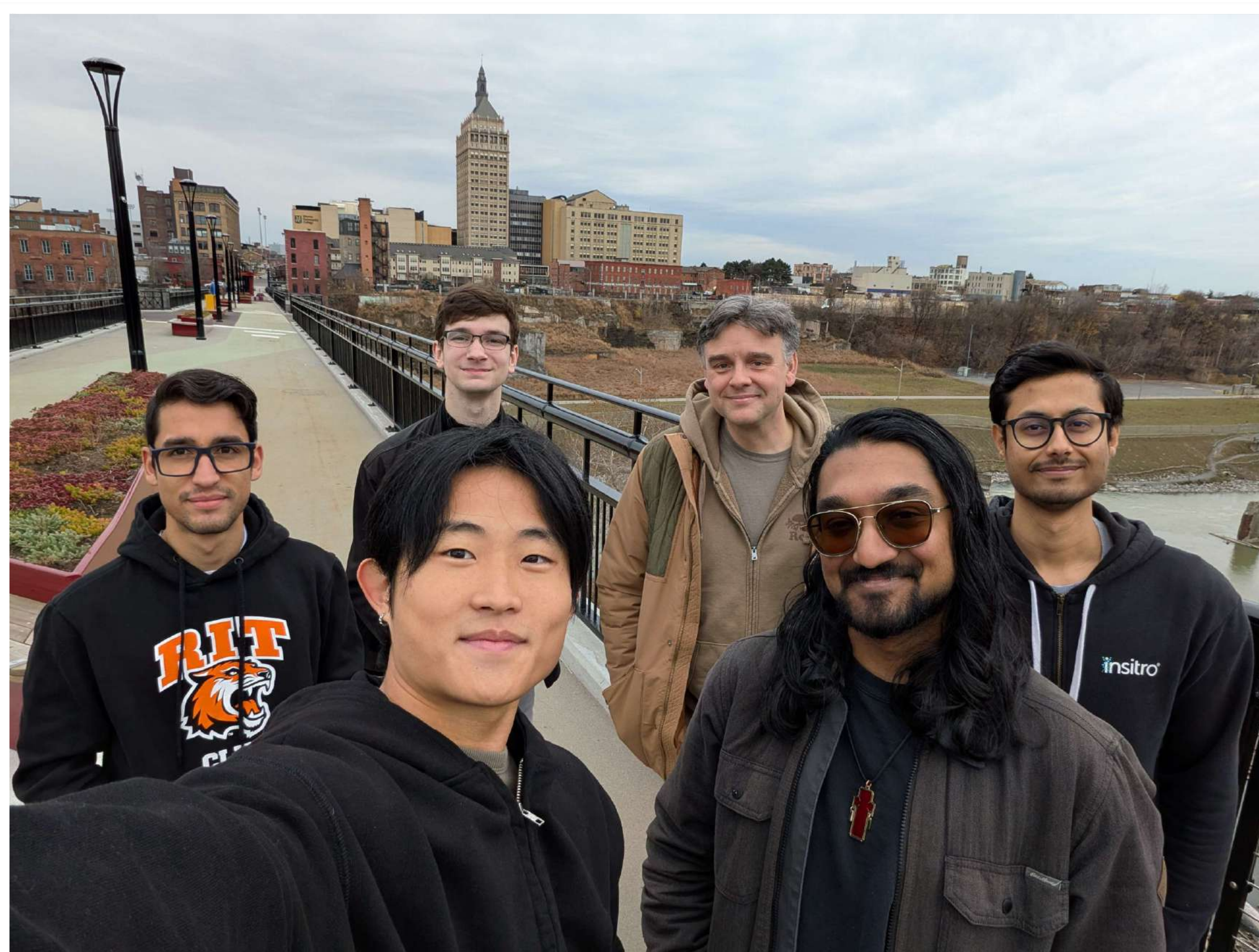
...however briefly.

- **Interpreting graphics requires context:** from within a document **and** from the reader's context (e.g., background knowledge). *Much is assumed.*
- **Graphics representations** impact both the types of information, and patterns in information that can be readily found in search
- **Well-designed information systems** (LLM, search, other) align well with human information needs & tasks, and provide affordances for them
- **'Task jar'** comprised of six task can characterize and relate information processing by people; the same tasks are implemented in systems
- A key role for **document analysis** in search is **annotating** sources and queries with additional information for use in search

Thank you



Big thanks to Josep, Dimos, Adria, Artemis, the CVC, and everyone else at this **great meeting**.



Mathematical Information Retrieval: Search and Question Answering

Richard Zanibbi, Behrooz Mansouri, and Anurag Agarwal

Foundations and Trends allowed the final PDF to be released for free! Link to PDF:

